



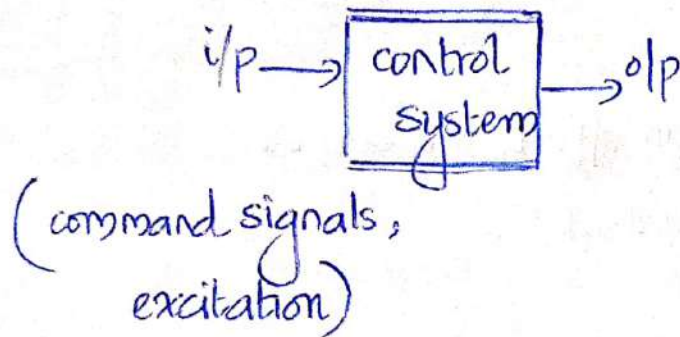
KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

Website: www.ktunotes.in

Control systems: A system consists of a no. of subsystems
 {Feedback present} - specific o/p, control the o/p



→ Types of Controls:

(i) ON-OFF control (switches)

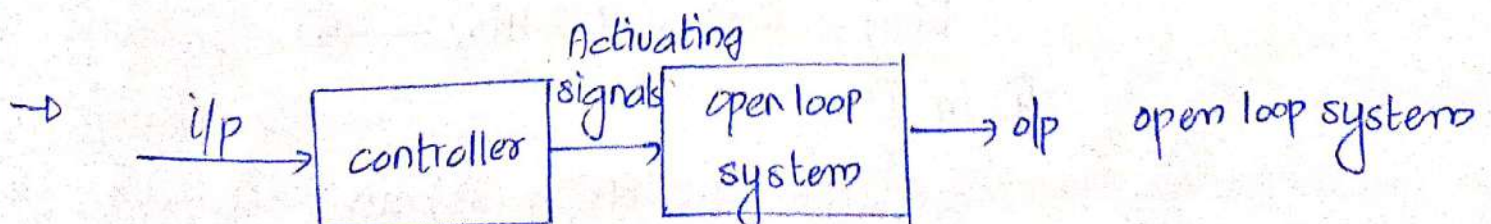
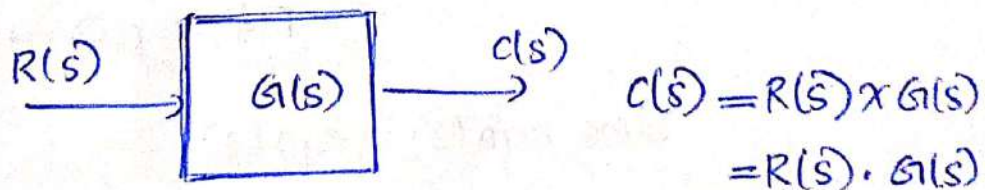
(ii) Discrete/continuous control (regulators in fans)

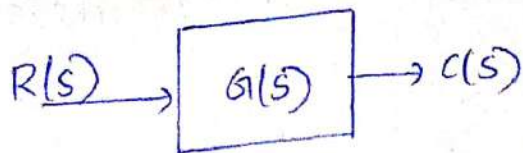
(iii) Manual/Automatic control (gear system, AC)

manual: filling of water tanks, flush system etc.

(iv) open loop system | closed loop system
 Eg: water tank. | feedback | Eg: automatic air conditioner.

→ General representation of system





$$C(s) = R(s) \times G(s)$$

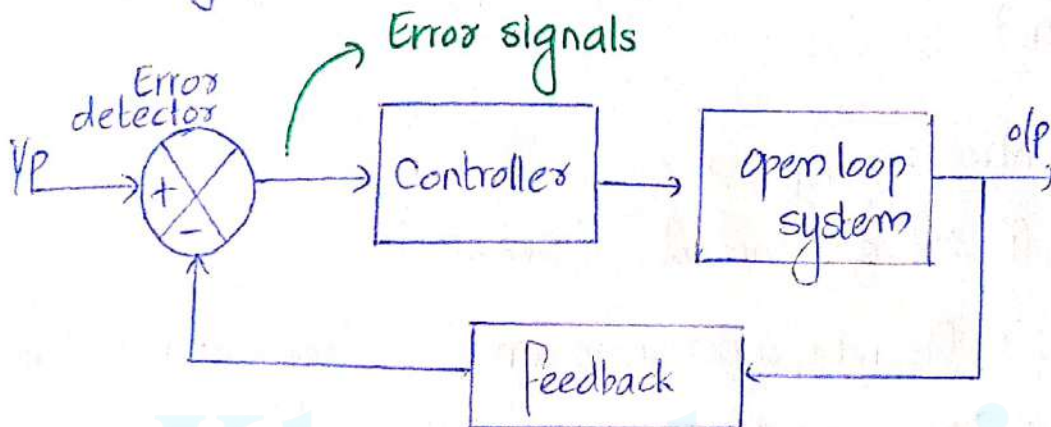
$$R(s) = \mathcal{L}\{r(t)\}$$

$$C(s) = \mathcal{L}\{c(t)\}$$

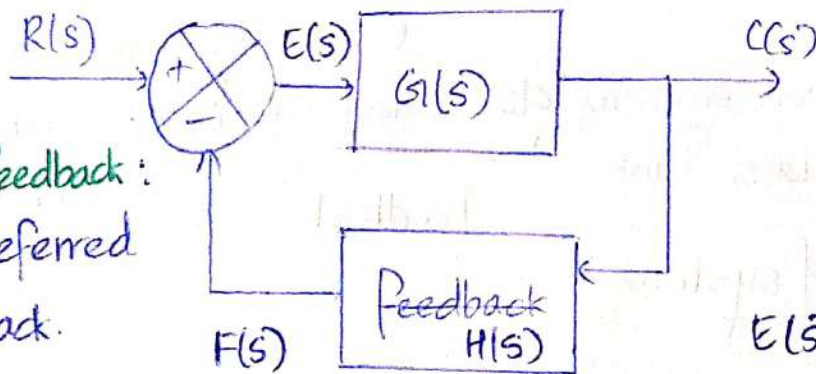
$$\frac{C(s)}{R(s)} = G(s) \text{ Transfer fn of the system}$$

closed loop system:

Drawback of open loop system: Noise can affect the output.
In closed loop system, we provide a feedback.



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$$C(s) = E(s) \cdot G(s) \text{ --- (1)}$$

General representation of a closed loop system

$$E(s) = R(s) - F(s) \text{ --- (2)}$$

$$F(s) = C(s) \cdot H(s) \text{ --- (3)}$$

Subs. eqn (3) in eqn (2);

$$E(s) = R(s) - C(s) \cdot H(s) \text{ --- (4)}$$

(4) in (1);

$$C(s) = (R(s) - C(s) \cdot H(s)) G(s)$$

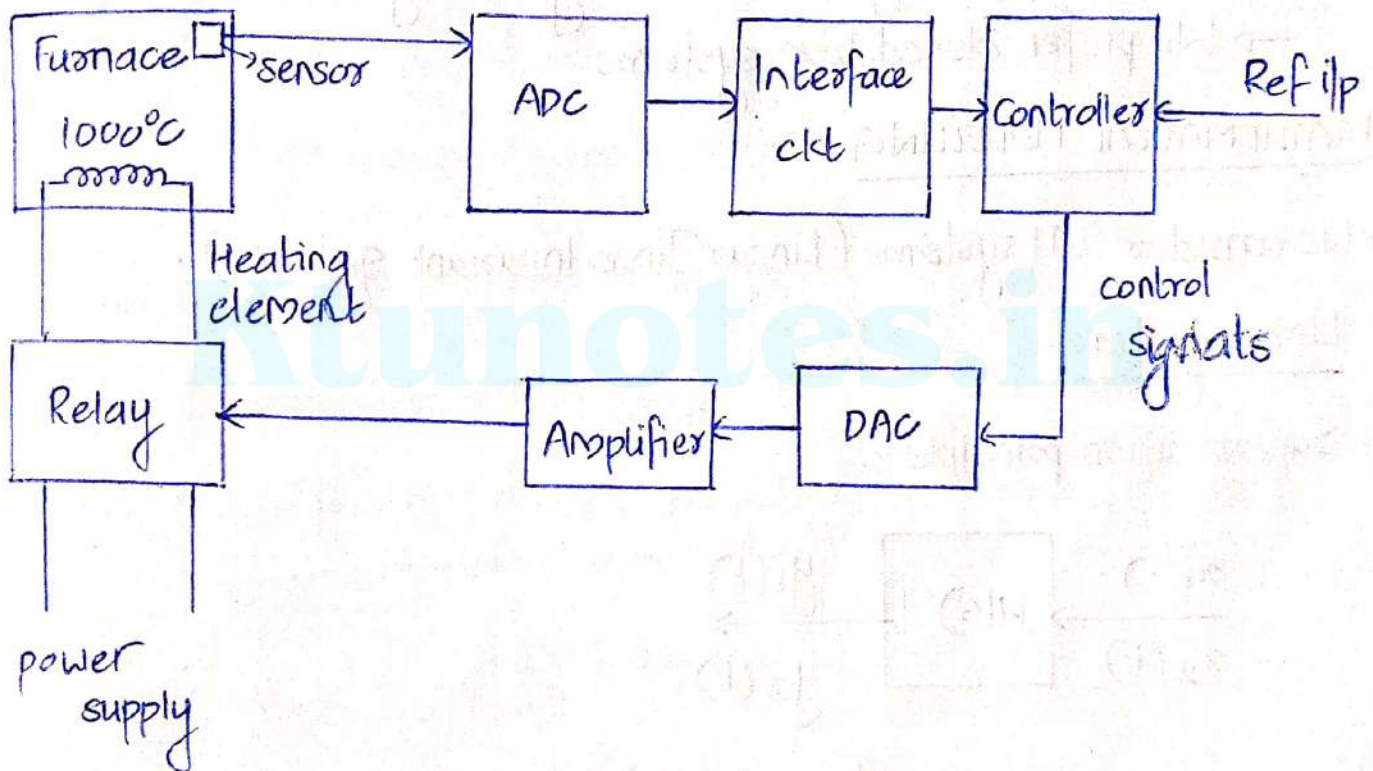
$$C(s) = R(s) \cdot G(s) - C(s) H(s) G(s)$$

$$C(s) [1 + H(s) G(s)] = R(s) G(s)$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s) H(s)} \quad \text{for -ve feedback.}$$

Note: Then for positive feedback, $\frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s) H(s)}$.

Example for C.L system



Advantages of open loop systems:

- Simple, Fast, easy to design
- cost effective

disadvantages:

- Cannot automatically compensate for external disturbances.
- less accuracy.

Advantages of closed loop system:

- Automatically compensate for ext disturbances (noise).
- Better accuracy.
- Manual control is minimized.
- More stable.

Disadvantages

- Complex to design.
- Cost is high.

Gain of OL system > Gain of CL system.

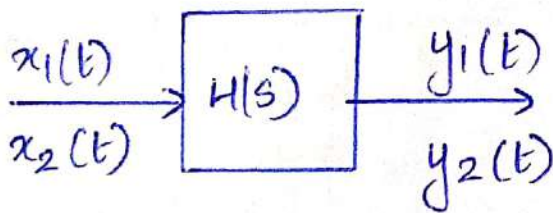
- We prefer closed loop systems.

MATHEMATICAL MODELLING

- We consider LTI systems (Linear Time Invariant systems).

Linear system

- Superposition principle.



$$\text{ip} \rightarrow ax_1(t) + bx_2(t).$$

$$\text{op} \rightarrow ay_1(t) + by_2(t)$$

$$y(t) = t \cdot x(t) \text{ --- ①}$$

$$y(t) = x^2(t) \text{ --- ②}$$

$$\text{①} \Rightarrow y_1(t) = t \cdot x_1(t)$$

$$y_2(t) = t \cdot x_2(t)$$

$$\text{ip} \rightarrow ax_1(t) + bx_2(t)$$

$$y_3(t) = t[ax_1(t) + bx_2(t)]$$

$$= tax_1(t) + tbx_2(t)$$

$$= \underline{ay_1(t) + by_2(t)} \quad \text{It is a linear system.}$$

$$(2) \Rightarrow y(t) = x^2(t)$$

$$y_1(t) = x_1^2(t)$$

$$y_2(t) = x_2^2(t) \quad \text{ip} \rightarrow x_1(t) + x_2(t)$$

$$y_3(t) = [x_1(t) + x_2(t)]^2$$

Non-linear system.

Time invariance

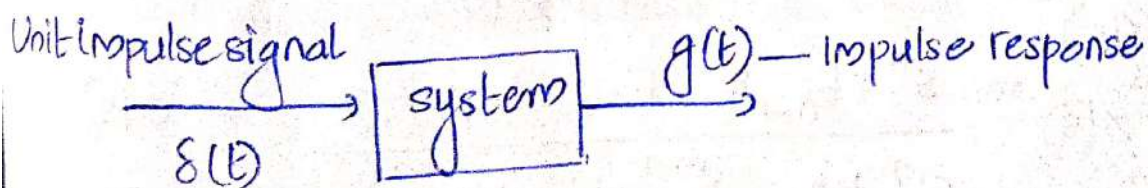
$$y(t) = t \cdot x(t) \rightarrow \text{① Time variant}$$

$$y(t) = 10x(t) \rightarrow \text{② Time invariant}$$

$$\text{①: } y'(t) = t x(t - t_0)$$

$$y''(t) = (t - t_0) x(t - t_0)$$

TRANSFER FUNCTION



Unit impulse, $\delta(t) = 0 \quad t \neq 0$
 $\neq 0 \quad t = 0$



$$G(s) = \frac{C(s)}{R(s)}$$

$G(s)$ is the LT of the impulse response $R(s)$

→ It is the Laplace transform of o/p signal | all initial conditions are set to zero.
Laplace transform of i/p signal

→ All initial conditions are set to 0.

→ classical way of representing the system in terms of differential eqn

$$\frac{d^m y}{dt^m} + a_{m-1} \frac{d^{m-1} y}{dt^{m-1}} + \dots + a_1 \frac{dy}{dt} + a_0$$

$$= \frac{d^n x}{dt^n} + b_{n-1} \frac{d^{n-1} x}{dt^{n-1}} + \dots + b_1 \frac{dx}{dt} + b_0$$

$$y(t) = \text{o/p}$$

$$x(t) = \text{i/p}$$

$$\mathcal{L}\left\{\frac{d}{dt}y(t)\right\} = sY(s) - y(0)$$

$$\mathcal{L}\left\{\frac{d^2}{dt^2}y(t)\right\} = s^2Y(s) - s \cdot y(0) - y'(0)$$

Initial conditions

$$\mathcal{L}\left\{\frac{d^3}{dt^3}y(t)\right\} = s^3Y(s) - s^2y(0) - sy'(0) - y''(0)$$

$$\longrightarrow s^m Y(s) + a_{m-1} s^{m-1} Y(s) + \dots + a_1 s Y(s) + a_0 Y(s)$$

$$= s^n X(s) + b_{n-1} s^{n-1} X(s) + \dots + b_1 s X(s) + b_0 X(s)$$

$$\longrightarrow \text{Transfer fn } \frac{Y(s)}{X(s)} = \frac{s^n + b_{n-1} s^{n-1} + \dots + b_1 s + b_0}{s^m + a_{m-1} s^{m-1} + \dots + a_1 s + a_0} = G(s)$$

Qn: Determine the transfer fn of a system governed by the differential

$$\text{eqn } \frac{d^3 y}{dt^3} + 5 \frac{d^2 y}{dt^2} + 10 \frac{dy}{dt} + 8y(t) = \frac{du}{dt} + 5u(t)$$

$$\mathcal{L}\left\{ \frac{du}{dt} + 5u(t) \right\} = \mathcal{L}\left\{ \frac{d^3 y}{dt^3} + 5 \frac{d^2 y}{dt^2} + 10 \frac{dy}{dt} + 8y(t) \right\}$$

$$\frac{Y(s)}{U(s)} = \frac{s+5}{s^3+5s^2+10s+8}$$

$$\text{Qn: } \frac{C(s)}{R(s)} = \frac{s^5+2s^2+5s+3}{s^5+8s^4+7s^3+5s^2+s+2}$$

$$s^5 C(s) + 8s^4 C(s) + 7s^3 C(s) + 5s^2 C(s) + s C(s) + 2C(s) = s^3 R(s) + 2s^2 R(s) + 5s R(s) + 3R(s)$$

$$\frac{d^5 c}{dt^5} + 8 \frac{d^4 c}{dt^4} + 7 \frac{d^3 c}{dt^3} + 5 \frac{d^2 c}{dt^2} + \frac{dc}{dt} + 2c(t) = \frac{d^3 r}{dt^3} + 2 \frac{d^2 r}{dt^2} + 5 \frac{dr}{dt} + 3r(t)$$

 Transfer fn can be defined as:

(1) LT of the impulse response | all initial conditions are set to zero

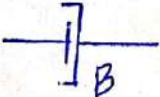
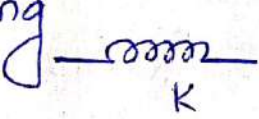
(2) $\frac{\text{LT of Dlp signal}}{\text{LT of ilp signal}}$ | all initial conditions are set to zero.

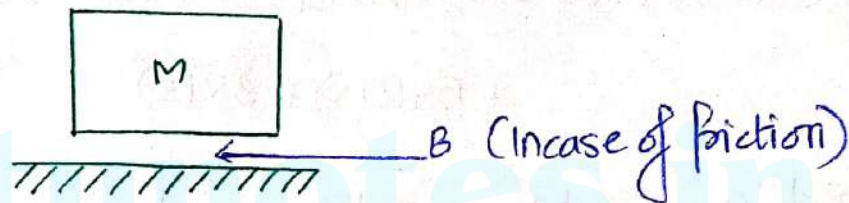
Properties of Transfer fn:

- Transfer fn is applicable to only LTI systems.
- All the initial conditions are set to zero.
- Transfer fn is independent of the i/p to the system.

Modelling of mechanical Systems

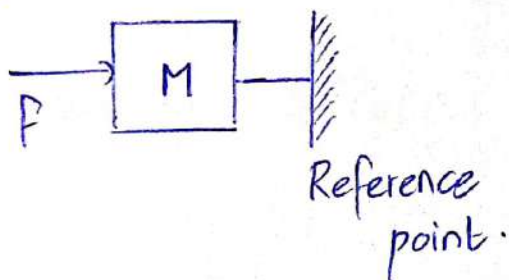
Basic features:

- Weight — Mass  M: mass
- Frictional force — dashpot  B: Viscous frictional Coefficient
- Elastic deformation — spring  K: stiffness constant of a spring



Force Balance Equations of ideal elements

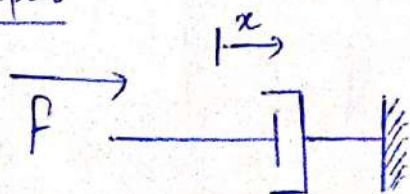
Mass x - displacement



Opposing force $F_m \propto \frac{d^2x}{dt^2}$

$$F = F_m = M \frac{d^2x}{dt^2}$$

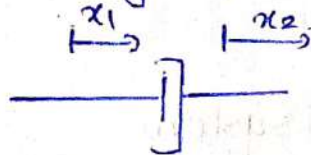
Dashpot



Opposing force due to the dashpot, $F_B \propto \frac{dx(t)}{dt}$

$$F = F_B = B \frac{dx}{dt}$$

case (ii) not attached to a reference point $\Rightarrow$ the displacement on either sides say, x_1 and x_2 .

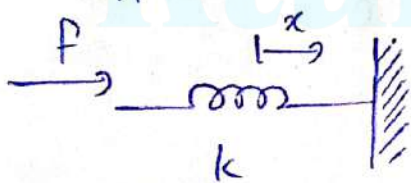


$$F_B \propto \frac{d(x_1 - x_2)}{dt}$$

$$F = F_B = B \frac{d(x_1 - x_2)}{dt}$$

Spring

Stiffness constant: k



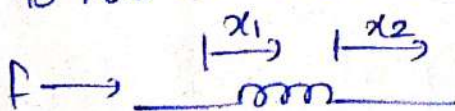
Opposing force $F_k \propto x$

$$F_k = kx$$

$$F = F_k = kx$$

$\swarrow$ applied force $\downarrow$ opposing force

case (ii) spring is not attached to the ref. point.



$$F_k \propto (x_1 - x_2)$$

$$F_k = k(x_1 - x_2)$$

$$\underline{\underline{F = F_k = k(x_1 - x_2)}}$$

Qn:

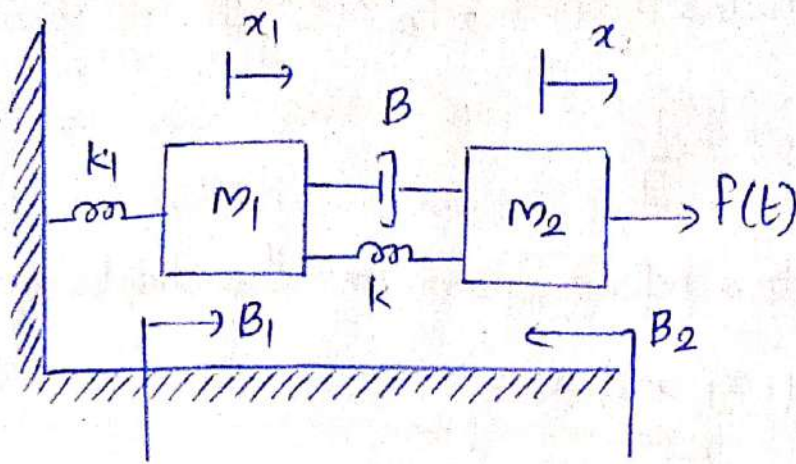


Fig: Mechanical translational system.

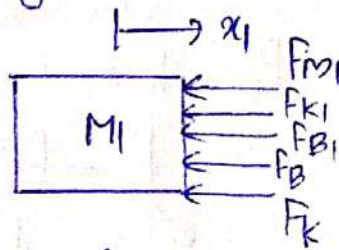
Find the transfer function $\frac{X(s)}{F(s)}$.

Soln: - Find the nodes in the system. The mass elements in the system can be considered as nodes.

$\therefore$ This system has 2 nodes.

Next step is to draw the free body diagram of each node.

Free Body diagram of M_1 :



$$F_{m1} = M_1 \frac{d^2 x_1}{dt^2}$$

$$F_{B1} = B_1 \frac{dx_1}{dt}$$

$$F_B = B \frac{d}{dt} (x_1 - x)$$

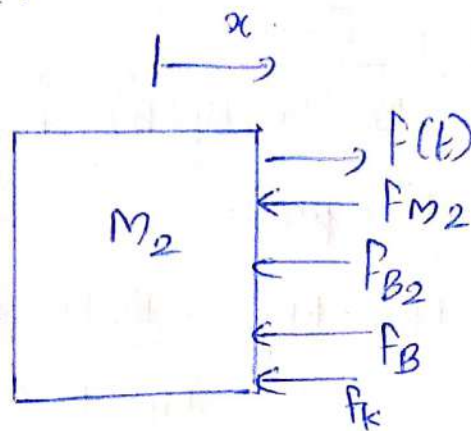
$$F_{k1} = k_1 x_1$$

$$F_k = k(x_1 - x)$$

$$M_1 \frac{d^2 x_1}{dt^2} + B_1 \frac{dx_1}{dt} + B \frac{d}{dt} (x_1 - x) + k_1 x_1 + k (x_1 - x) = 0 \quad \text{--- (1)}$$

This is the diff. eqn associated with mass M_1 (node).

Freebody diagram of M_2 :



$$F_{M_2} = M_2 \frac{d^2 x}{dt^2}$$

$$F_{B_2} = B_2 \frac{dx}{dt}$$

$$F_B = B \frac{d}{dt} (x - x_1)$$

$$F_k = k (x - x_1)$$

The diff. eqn is $M_2 \frac{d^2 x}{dt^2} + B_2 \frac{dx}{dt} + B \frac{d}{dt} (x - x_1) + k (x - x_1) = F(t) \quad \text{--- (2)}$

$$\textcircled{1} \Rightarrow M_1 s^2 x_1(s) + B_1 s x_1(s) + B s [x_1(s) - x(s)] + k_1 x_1(s) + k [x_1(s) - x(s)] = 0 \quad \text{--- (3)}$$

$$\textcircled{2} \Rightarrow M_2 s^2 x(s) + B_2 s x(s) + B x(s) s [x(s) - x_1(s)] + k x(s) [x(s) - x_1(s)] = F(s) \quad \text{--- (5)}$$

③ can be simplified as:

$$x_1(s) [M_1 s^2 + (B_1 + B) s + (k_1 + k)] - x(s) [B s + k] = 0 \quad \text{--- (4)}$$

⑤ can be simplified as:

$$X(s) [M_2 s^2 + (B_2 + B)s + k] - X_1(s) [Bs + k] = F(s) \quad \text{--- ⑥}$$

From eqn ④,

$$X_1(s) = X(s) \cdot \frac{Bs + k}{M_1 s^2 + (B_1 + B)s + (k_1 + k)}$$

Subs value of $X_1(s)$ in eqn ⑥,

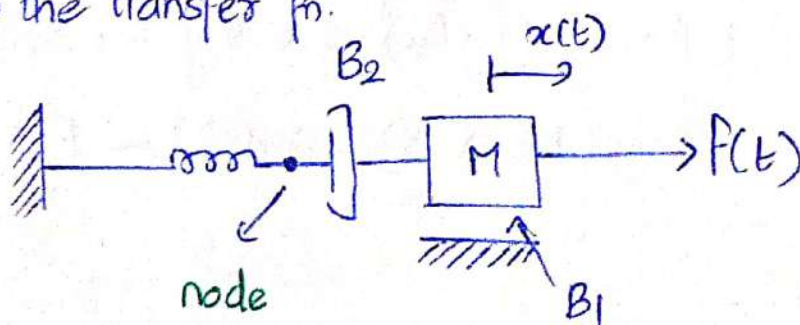
$$X(s) [M_2 s^2 + (B_2 + B)s + k] - \frac{(Bs + k)(Bs + k) X(s)}{M_1 s^2 + (B_1 + B)s + (k_1 + k)} = F(s)$$

$$X(s) \left[M_2 s^2 + (B_2 + B)s + k - \frac{(Bs + k)^2}{M_1 s^2 + (B_1 + B)s + (k_1 + k)} \right] = F(s)$$

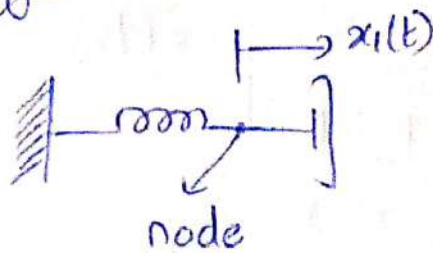
$$X(s) \left[\frac{(M_2 s^2 + (B_2 + B)s + k)(M_1 s^2 + (B_1 + B)s + (k_1 + k)) - (Bs + k)^2}{M_1 s^2 + (B_1 + B)s + (k_1 + k)} \right] = F(s)$$

$$\frac{X(s)}{F(s)} = \frac{M_1 s^2 + (B_1 + B)s + (k_1 + k)}{(M_2 s^2 + (B_2 + B)s + k)(M_1 s^2 + (B_1 + B)s + (k_1 + k)) - (Bs + k)^2}$$

Qn: Formulate the diff eqns of the mechanical translational system and obtain the transfer fn.

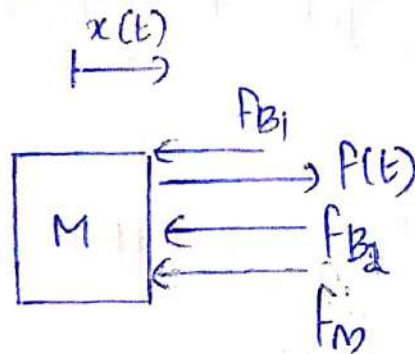


Freebody diagram for node M



Transfer fn $\rightarrow \frac{x(s)}{F(s)}$

Freebody diagram for node M

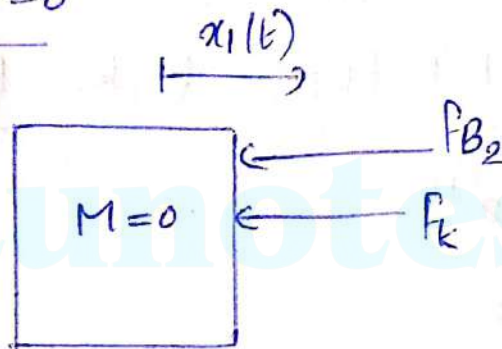


$$F_{B1} = B_1 \frac{dx}{dt}$$

$$F_{B2} = B_2 \frac{d(x - x_1)}{dt}$$

$$F_M = M \frac{d^2x}{dt^2}$$

FBD for node M=0



$$F_{B2} = B_2 \frac{d(x_1 - x)}{dt}$$

$$F_k = kx_1$$

diff eqn is $M \frac{d^2x}{dt^2} + B_1 \frac{dx}{dt} + B_2 \frac{d(x - x_1)}{dt} = F(t)$

$$Ms^2x(s) + B_1sX(s) + B_2s[X(s) - x_1(s)] = F(s) \text{ --- (1)}$$

$$B_2 \frac{d(x_1 - x)}{dt} + kx_1 = 0$$

$$B_2s[x_1(s) - x(s)] + kx_1(s) = 0 \text{ --- (2)}$$

$$B_2x_1(s) - B_2x(s) + kx_1(s) = 0$$

$$(B_2s + k)x_1(s) = B_2x(s) \quad x_1(s) = \frac{B_2x(s)s}{k + B_2s} \text{ --- (3)}$$

Subs (3) in (1),

$$Ms^2x(s) + B_1sX(s) + B_2s \left[X(s) - \frac{B_2X(s)s}{k+B_2s} \right] = F(s)$$

$$Ms^2x(s) + B_1sX(s) + B_2X(s)s - \frac{B_2^2x(s)s}{k+B_2s} = F(s)$$

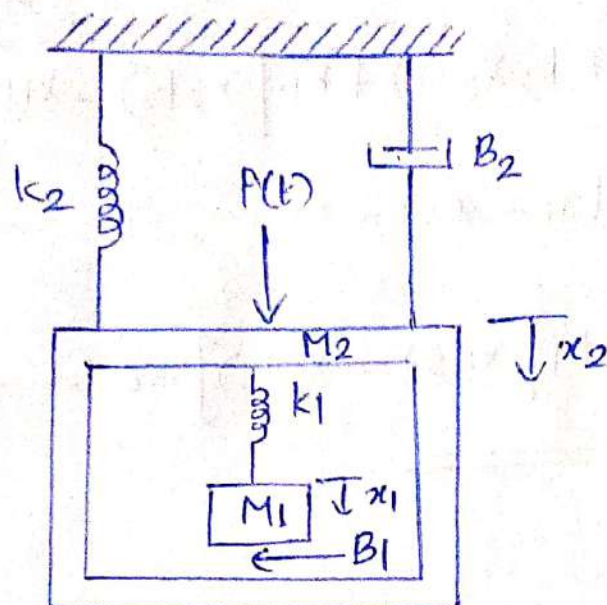
$$\frac{[Ms^2x(s) + B_1sX(s) + B_2sX(s)] [k+B_2s] - B_2^2x(s)s^2}{k+B_2s} = F(s)$$

$$\frac{Ms^2kX(s) + MB_2s^3X(s) + B_1skX(s) + B_1B_2s^2X(s) + B_2kX(s)s + B_2^2s^2X(s) - B_2^2X(s)s^2}{k+B_2(s)}$$

$$\rightarrow \frac{X(s) [Ms^2 + B_1s + B_2s] [k+B_2s] - B_2^2s^2}{k+B_2(s)} = F(s)$$

$$\frac{X(s)}{F(s)} = \frac{k+B_2(s)}{(Ms^2 + (B_1+B_2)s)(k+B_2s) - B_2^2s^2}$$

Qn: Formulate the diff eqns for the mechanical systems:



Freebody diagram of M_2

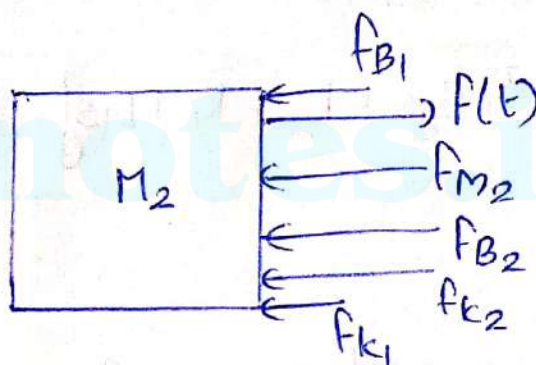
$$f_{M_2} = M_2 \frac{d^2 x_2}{dt^2}$$

$$f_{B_2} = B_2 \frac{dx_2}{dt}$$

$$f_{k_2} = k_2 (x_2 - x_1)$$

$$f_{k_1} = k_1 (x_2 - x_1)$$

$$f_{B_1} = B_1 \frac{d(x_2 - x_1)}{dt}$$

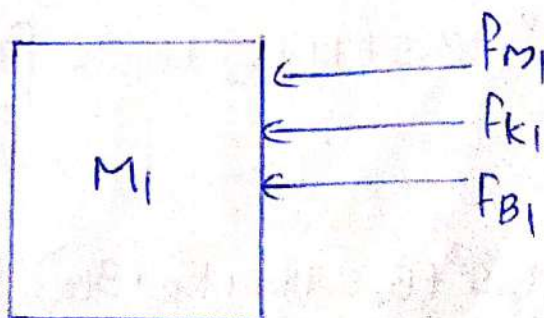


Freebody diagram of M_1

$$f_{M_1} = M_1 \frac{d^2 x_1}{dt^2}$$

$$f_{k_1} = k_1 (x_1 - x_2)$$

$$f_{B_1} = B_1 \frac{d(x_1 - x_2)}{dt}$$



$$M_2 \frac{d^2 x_2}{dt^2} + B_2 \frac{dx_2}{dt} + k_2 x_2 + k_1 (x_2 - x_1) + B_1 \frac{d(x_2 - x_1)}{dt} = F(t)$$

$$M_2 s^2 x_2(s) + B_2 x_2(s) \cdot s + k_2 x_2(s) + k_1 [x_2(s) - x_1(s)] + B_1 s [x_2(s) - x_1(s)] = F(s) \quad \text{--- ①}$$

$$M_1 \frac{d^2 x_1}{dt^2} + B_1 \frac{d(x_1 - x_2)}{dt} + k_1 (x_1 - x_2) = 0$$

$$M_1 s^2 x_1(s) + B_1 s [x_1(s) - x_2(s)] + k_1 [x_1(s) - x_2(s)] = 0 \quad \text{--- ②}$$

From eqn ②,

$$x_1(s) [M_1 s^2 + B_1 s + k_1] - x_2(s) [B_1 s + k_1] = 0 \quad \text{--- ③}$$

From eqn ①,

$$x_2(s) [M_2 s^2 + B_2 s + k_2 + k_1 + B_1 s] - x_1(s) [B_1 s + k_1] = F(s) \quad \text{--- ④}$$

From eqn ③;

$$x_1(s) = \frac{x_2(s) [B_1 s + k_1]}{[M_1 s^2 + B_1 s + k_1]} \quad \text{--- ⑤}$$

Substituting ⑤ in ④,

$$x_2(s) [M_2 s^2 + B_2 s + k_1 + k_2 + B_1 s] - \frac{[B_1 s + k_1] x_2(s) [B_1 s + k_1]}{M_1 s^2 + B_1 s + k_1} = F(s)$$

$$x_2(s) = \left[\frac{(M_2 s^2 + B_2 s + k_1 + k_2 + B_1 s)(M_1 s^2 + B_1 s + k_1) - (B_1 s + k_1)^2}{M_1 s^2 + B_1 s + k_1} \right] F(s)$$

$$\frac{X_2(s)}{F(s)} = \frac{M_1 s^2 + B_1 s + k_1}{(M_2 s^2 + B_2 s + k_1 + k_2 + B_1 s)(M_1 s^2 + B_1 s + k_1) - (B_1 s + k_1)^2}$$

From eqn (3),

$$X_2(s) = \frac{X_1(s) [M_1 s^2 + B_1 s + k_1]}{B_1 s + k_1} \quad \text{--- (6)}$$

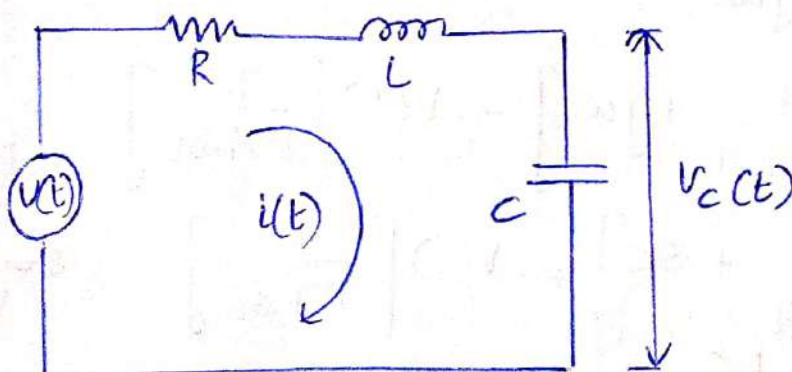
Substituting (6) in (4);

$$\frac{X_1(s) [M_1 s^2 + B_1 s + k_1]}{B_1 s + k_1} [M_2 s^2 + B_2 s + k_1 + k_2 + B_1 s] - X_1(s) [B_1 s + k_1] = F(s)$$

$$X_1(s) \left[\frac{(M_1 s^2 + B_1 s + k_1)(M_2 s^2 + B_2 s + k_1 + k_2 + B_1 s) - (B_1 s + k_1)^2}{B_1 s + k_1} \right] = F(s)$$

$$\frac{X_1(s)}{F(s)} = \frac{B_1 s + k_1}{(M_1 s^2 + B_1 s + k_1)(M_2 s^2 + B_2 s + k_1 + k_2 + B_1 s) - (B_1 s + k_1)^2}$$

ELECTRICAL SYSTEMS



$$V(t) = Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i dt \quad \text{--- (1)}$$

$$V(s) = I(s)R + LS I(s) + \frac{1}{CS} I(s) \quad \text{--- (2)}$$

$$i(t) = C \frac{dv_c}{dt} \quad I(s) = CS V_c(s) \quad \text{--- (3)}$$

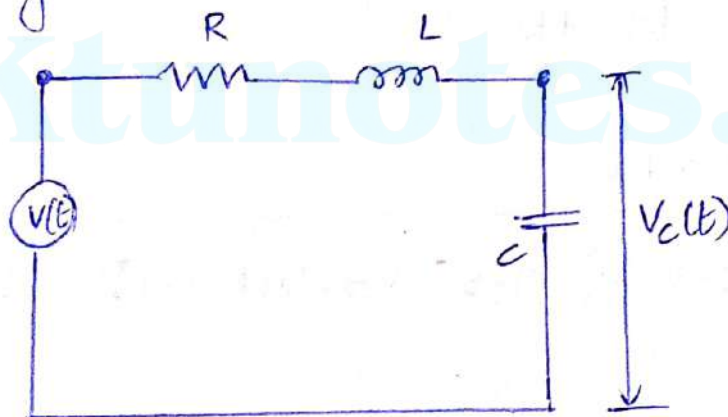
$$\text{So } V(s) = (CS V_c(s))R + LS(CS V_c(s)) + \frac{1}{CS}(CS V_c(s))$$

$$V(s) = RCS V_c(s) + LCS^2 V_c(s) + V_c(s)$$

$$V(s) = V_c(s) [RCS + LCS^2 + 1]$$

$$\frac{V_c(s)}{V(s)} = \frac{1}{RCS + LCS^2 + 1} \quad \text{--- (4)}$$

By nodal analysis:



$$\frac{V_c(t) - V(t)}{R + j\omega L} + \frac{V_c(t)}{1/j\omega C} = 0$$

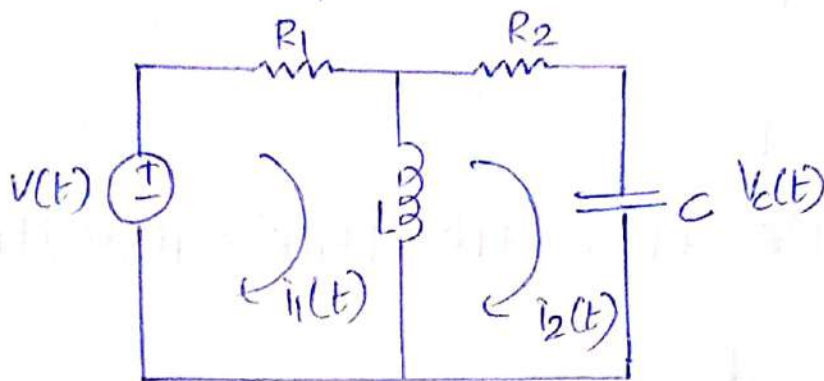
$$V_c(t) \left[\frac{1}{R + j\omega L} + j\omega C \right] = V(t) \left[\frac{1}{R + j\omega L} \right]$$

$$V_c(s) \left[\frac{1}{R + sL} + sC \right] = V(s) \left[\frac{1}{R + sL} \right] \quad s = j\omega$$

$$\frac{V_c(s)}{V(s)} = \frac{\left(\frac{1}{R + sL} \right)}{\left(\frac{1}{R + sL} + sC \right)}$$

$$= \frac{\left(\frac{1}{R+SL} \right)}{\left(\frac{1+SC(R+SL)}{R+SL} \right)} = \frac{1}{1+RCS+LCS^2}$$

Qn: Find the transfer fn $\frac{I_2(s)}{V(s)}$



Mesh analysis:

Mesh 1: $R_1 \frac{di_1(t)}{dt} + L \frac{d(i_1(t) - i_2(t))}{dt} = V(t)$

$$V(s) = R_1 I_1(s) + LS I_1(s) - LS I_2(s) \quad \text{--- ①}$$

Mesh 2: $L \left[\frac{di_2(t)}{dt} - \frac{di_1(t)}{dt} \right] + R_2 i_2(t) + \frac{1}{C} \int i_2(t) dt = 0$

$$LS I_2(s) - LS I_1(s) + R_2 I_2(s) + \frac{1}{CS} I_2(s) = 0 \quad \text{--- ②}$$

$$I_2(s) \left[LS + R_2 + \frac{1}{CS} \right] - LS I_1(s) = 0$$

$$I_1(s) = \frac{I_2(s) \left[LS + R_2 + \frac{1}{CS} \right]}{LS}$$

$$= \frac{I_2(s) (LCS^2 + R_2 CS + 1)}{LCS^2} \quad \text{--- ③}$$

Subs (3) in (1),

$$\textcircled{1} \Rightarrow V(s) = I_1(s) [R_1 + Ls] - LsI_2(s)$$

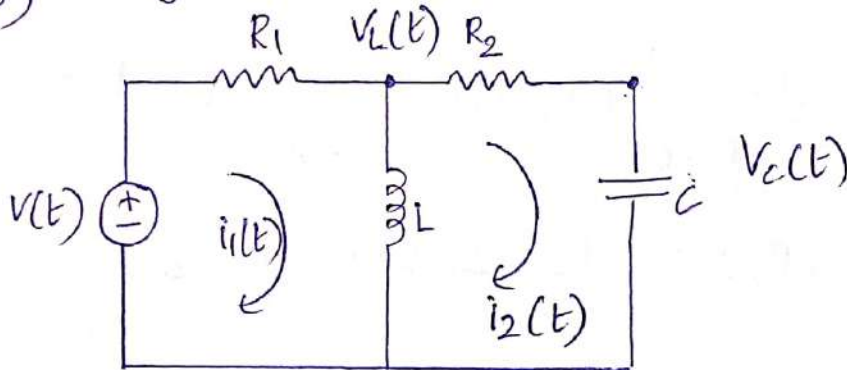
$$= \frac{I_2(s)}{Lcs^2} (Lcs^2 + R_2cs + 1)(R_1 + Ls) - LsI_2(s)$$

$$V(s) = \frac{I_2(s)}{Lcs^2} \left[(Lcs^2 + R_2cs + 1)(R_1 + Ls) - L^2cs^3 \right]$$

$$\frac{I_2(s)}{V(s)} = \frac{Lcs^2}{L^2cs^3 + LR_2cs^2 + Ls + LR_1cs^2 + R_1R_2cs + R_1 - L^2cs^3}$$

$$= \frac{Lcs^2}{LR_2cs^2 + Ls + LR_1cs^2 + R_1R_2cs + R_1}$$

(ii) $\frac{V_L(s)}{V(s)}$ using nodal analysis.



$$\frac{V_L(t) - V(t)}{R_1} + \frac{V_L(t)}{j\omega L} + \frac{V_L(t) - V_C(t)}{R_2 + \cancel{j\omega C}} = 0$$

$$\frac{V_L(s) - V(s)}{R_1} + \frac{V_L(s)}{Ls} + \frac{V_L(s) - V_C(s)}{R_2 + \cancel{\frac{1}{sC}}} = 0$$

$$V_L(s) \left[\frac{1}{R_1} + \frac{1}{Ls} + \frac{1}{R_2 + \frac{1}{sc}} \right] - \frac{V(s)}{R_1} - \frac{V_C(s)}{R_2 + \frac{1}{sc}} = 0 \quad \text{--- ①}$$

2nd node:

$$\frac{V_C(t) - V_L(t)}{R_2} + \frac{V_C(t)}{\frac{1}{j\omega c}} = 0$$

$$\frac{V_C(s) - V_L(s)}{R_2} + \frac{V_C(s)}{\frac{1}{sc}} = 0$$

$$V_C(s) \left[\frac{1}{R_2} + \frac{1}{\frac{1}{sc}} \right] - V_L(s) \left[\frac{1}{R_2} \right] = 0$$

$$V_L(s) = \frac{V_C(s) \times R_2 \left(\frac{1}{R_2} + sc \right)}{1}$$

$$V_L(s) = R_2 V_C(s) \left(\frac{1}{R_2} + sc \right) \quad \text{--- ②}$$

Substituting ② in ①,

$$R_2 V_C(s) \left(\frac{1}{R_2} + sc \right) \left(\frac{1}{R_1} + \frac{1}{Ls} + \frac{1}{R_2} \right) - \frac{V(s)}{R_1} - \frac{V_C(s)}{R_2 + \frac{1}{sc}} = 0$$

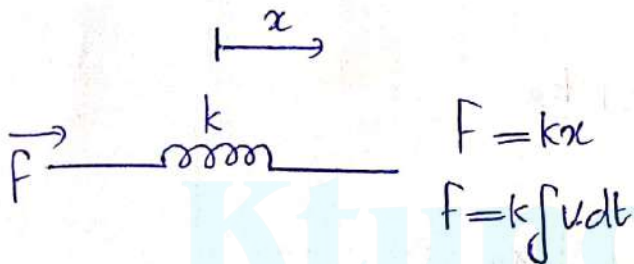
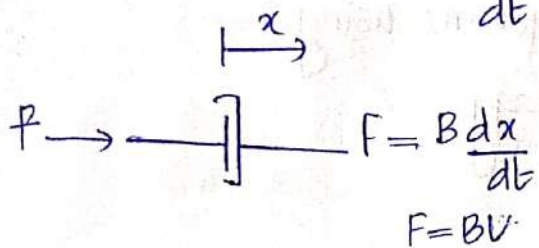
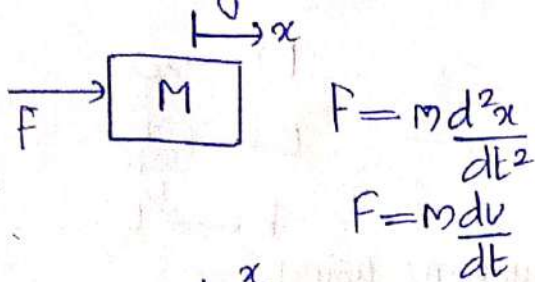
$$V_C(s) \left[(1 + R_2 cs) \left(\frac{1}{R_1} + \frac{1}{Ls} + \frac{1}{R_2} - \frac{1}{(R_2 + \frac{1}{sc})} \right) \right] = \frac{V(s)}{R_1}$$

$$\frac{V_C(s)}{V(s)} = \frac{1}{R_1 \left[(1 + R_2 cs) \left(\frac{1}{R_1} + \frac{1}{Ls} + \frac{1}{R_2} - \frac{1}{(R_2 + \frac{1}{sc})} \right) \right]}$$

FORCE-VOLTAGE ANALOGY

MECHANICAL

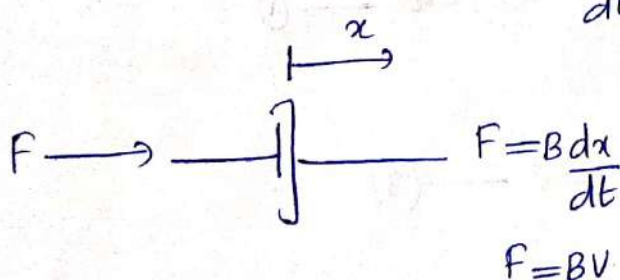
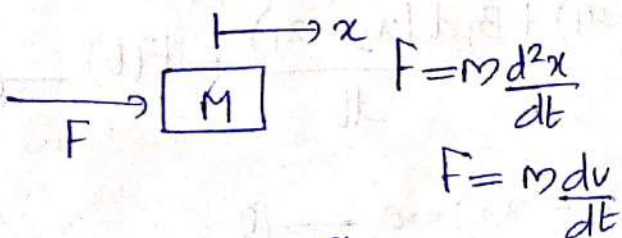
Input : Force (F)
output : Velocity (v)



FORCE-CURRENT ANALOGY

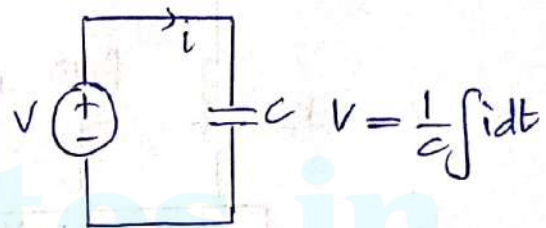
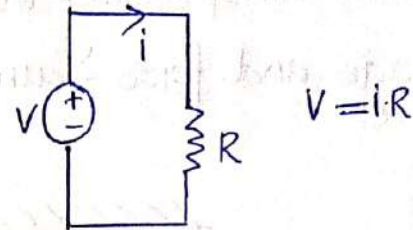
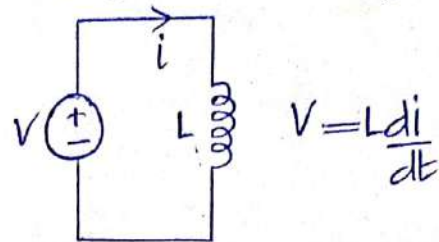
MECHANICAL

input : Force (F)
output : velocity (v)



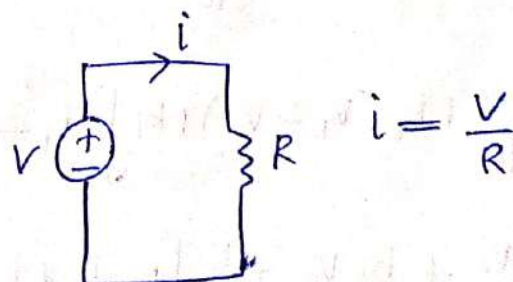
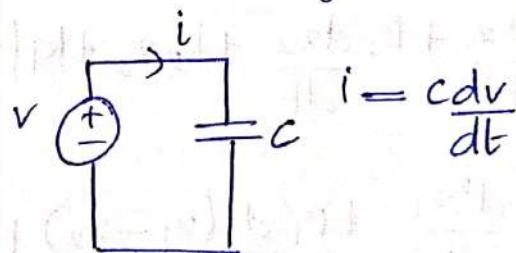
ELECTRICAL

Input : voltage (v)
output : current (i)

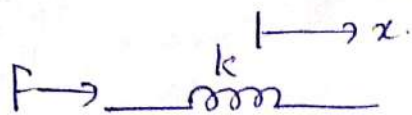


ELECTRICAL

Input : current (i)
output : voltage (v)

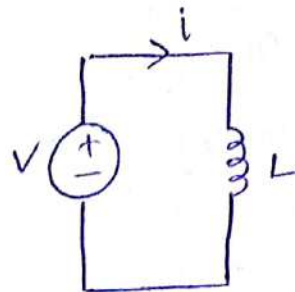


$M \leftrightarrow L$
 $B \leftrightarrow R$
 $k \leftrightarrow 1/C$



$$F = kx$$

$$F = k \int v dt$$



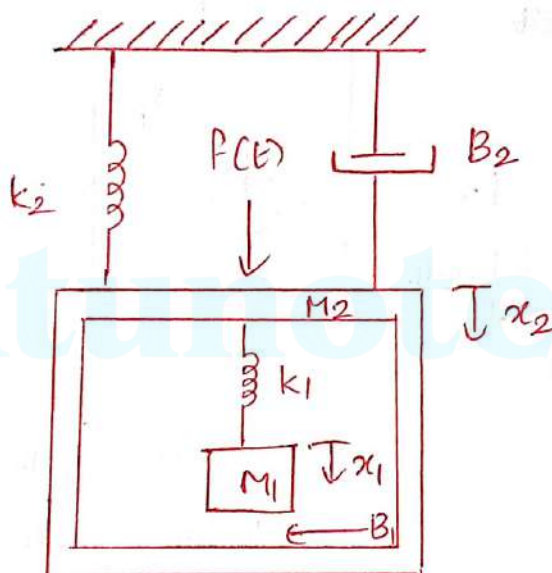
$$i = \frac{1}{L} \int v dt$$

$$M \leftrightarrow C$$

$$B \leftrightarrow \frac{1}{R}$$

$$k \leftrightarrow \frac{1}{L}$$

2: Convert into corresponding electrical system using force-voltage and force-current analogy.



$$M_2 \frac{d^2 x_2}{dt^2} + B_2 \frac{dx_2}{dt} + k_2 x_2 + k_1 (x_2 - x_1) + B_1 \frac{d(x_2 - x_1)}{dt} = F(t) \quad \text{--- (2)}$$

$$M_1 \frac{d^2 x_1}{dt^2} + B_1 \frac{d(x_1 - x_2)}{dt} + k_1 (x_1 - x_2) = 0 \quad \text{--- (1)}$$

$$M_1 \frac{dv_1}{dt} + B_1 (v_1 - v_2) + k_1 \int (v_1 - v_2) dt = 0 \quad \text{--- (3)}$$

$$M_2 \frac{dv_2}{dt} + B_2 v_2 + k_2 \int v_2 dt + k_1 \int (v_2 - v_1) dt + B_1 (v_2 - v_1) = F(t) \quad \text{--- (4)}$$

Force-voltage analogy

$$M_1 \leftrightarrow L_1$$

$$B_1 \leftrightarrow R_1$$

$$k_1 \leftrightarrow \frac{1}{C_1}$$

$$M_2 \leftrightarrow L_2$$

$$B_2 \leftrightarrow R_2$$

$$k_2 \leftrightarrow \frac{1}{C_2}$$

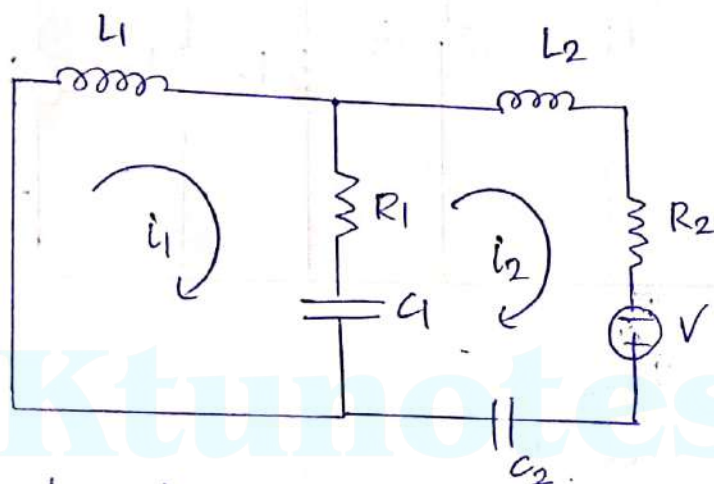
$$\text{Force (F)} \leftrightarrow \text{voltage (V)}$$

$$\text{Velocity (v)} \leftrightarrow \text{current (i)}$$

$$v_2 \leftrightarrow i_2$$

$$L_1 \frac{di_1}{dt} + R_1 (i_1 - i_2) + \frac{1}{C_1} \int (i_1 - i_2) dt = 0$$

$$L_2 \frac{di_2}{dt} + R_2 i_2 + \frac{1}{C_2} \int i_2 dt + \frac{1}{C_1} \int (i_2 - i_1) dt + R_1 (i_2 - i_1) = V(t)$$



Force-current analogy

$$M_1 \leftrightarrow C_1$$

$$M_2 \leftrightarrow C_2$$

$$B_1 \leftrightarrow \frac{1}{R_1}$$

$$B_2 \leftrightarrow \frac{1}{R_2}$$

$$k_1 \leftrightarrow \frac{1}{L_1}$$

$$k_2 \leftrightarrow \frac{1}{L_2}$$

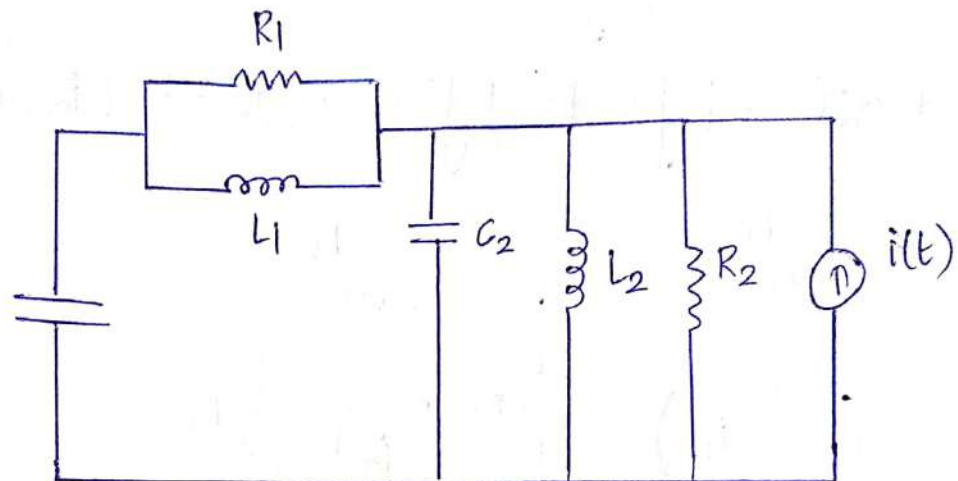
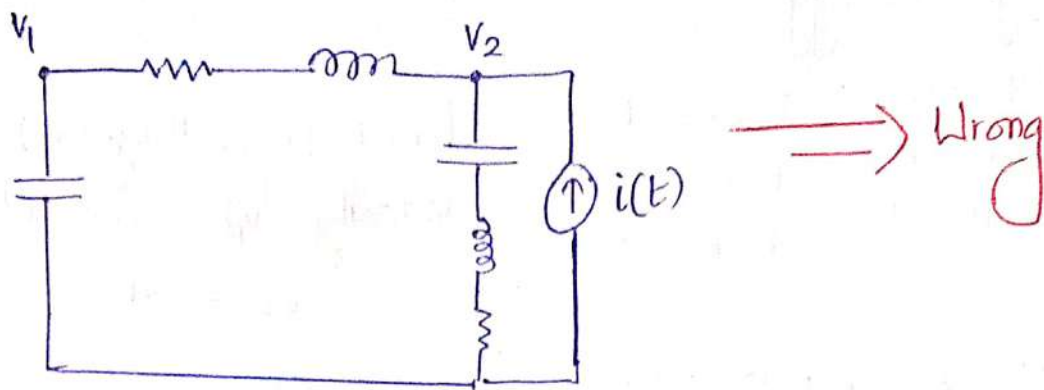
$$F(t) \leftrightarrow i$$

$$\text{Velocity (v)} \leftrightarrow V$$

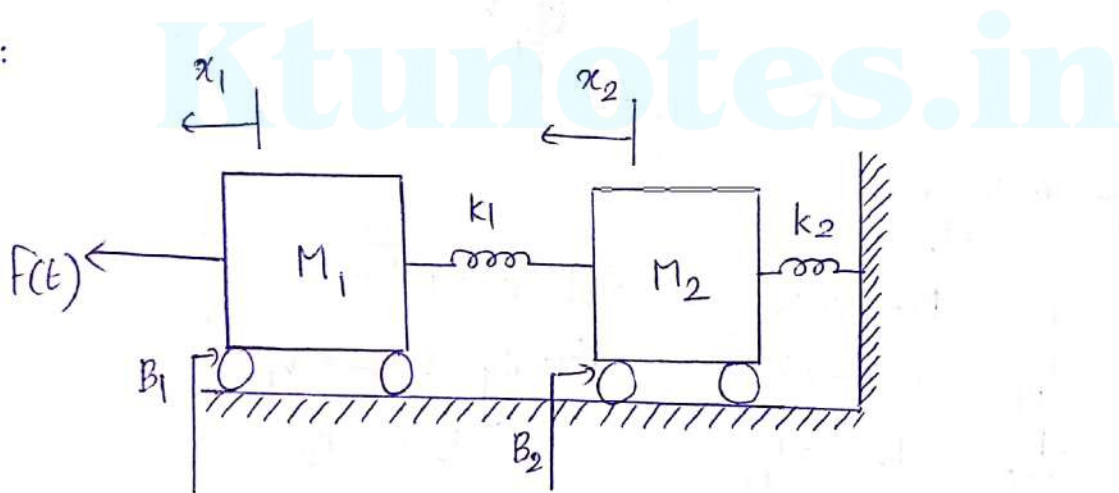
$$v_2 \leftrightarrow V_2$$

$$C_1 \frac{dV_1}{dt} + \frac{1}{R_1} (V_1 - V_2) + \frac{1}{L_1} \int (V_1 - V_2) dt = 0$$

$$C_2 \frac{dV_2}{dt} + \frac{1}{R_2} V_2 + \frac{1}{L_2} \int V_2 dt + \frac{1}{L_1} \int (V_2 - V_1) dt + \frac{1}{R_1} (V_2 - V_1) = i(t)$$



Qn:

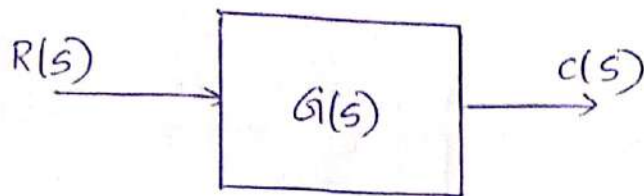


Find the transfer fn $\frac{x_2(s)}{F(s)}$ for the mechanical system

Also draw the force-voltage and force-current analogy

(Tutorial) 54/12/2021

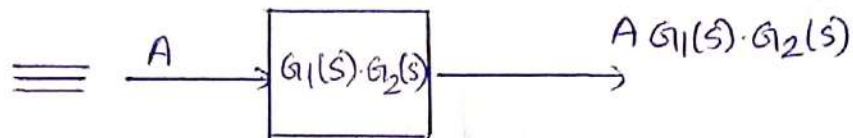
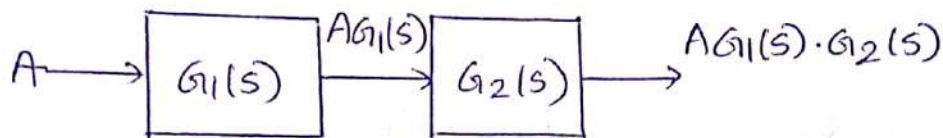
BLOCK DIAGRAM REDUCTION METHOD



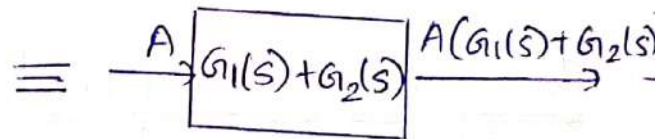
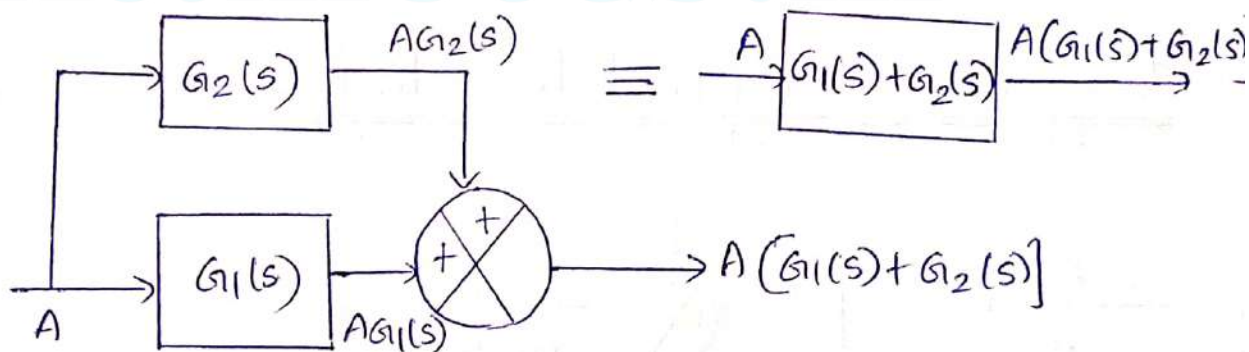
$$G(s) = \frac{C(s)}{R(s)}$$

Rules of block diagram reduction:

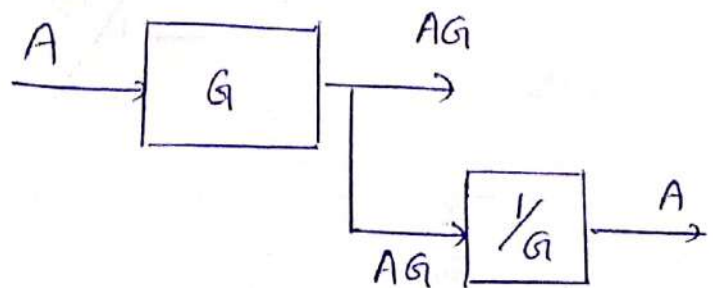
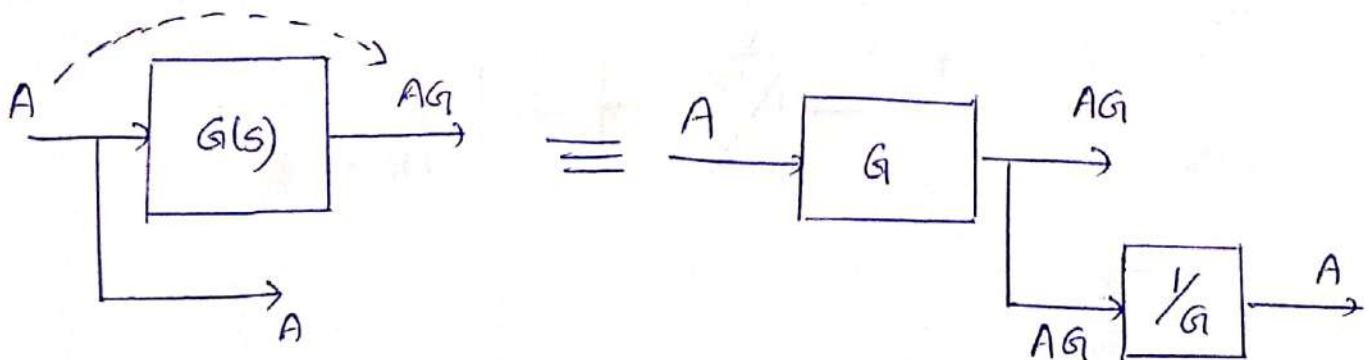
1. Blocks in cascade



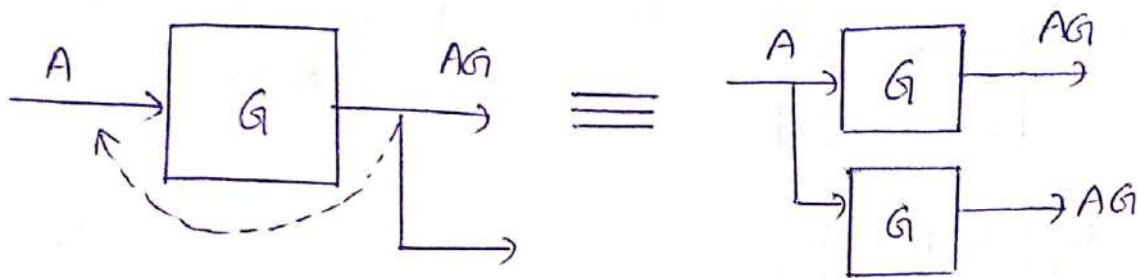
2. Blocks in parallel



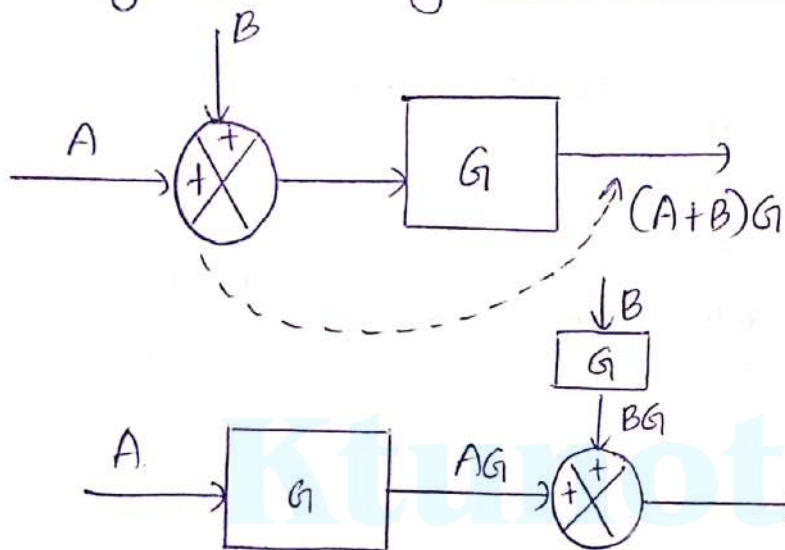
3. Moving branch point ahead of the block



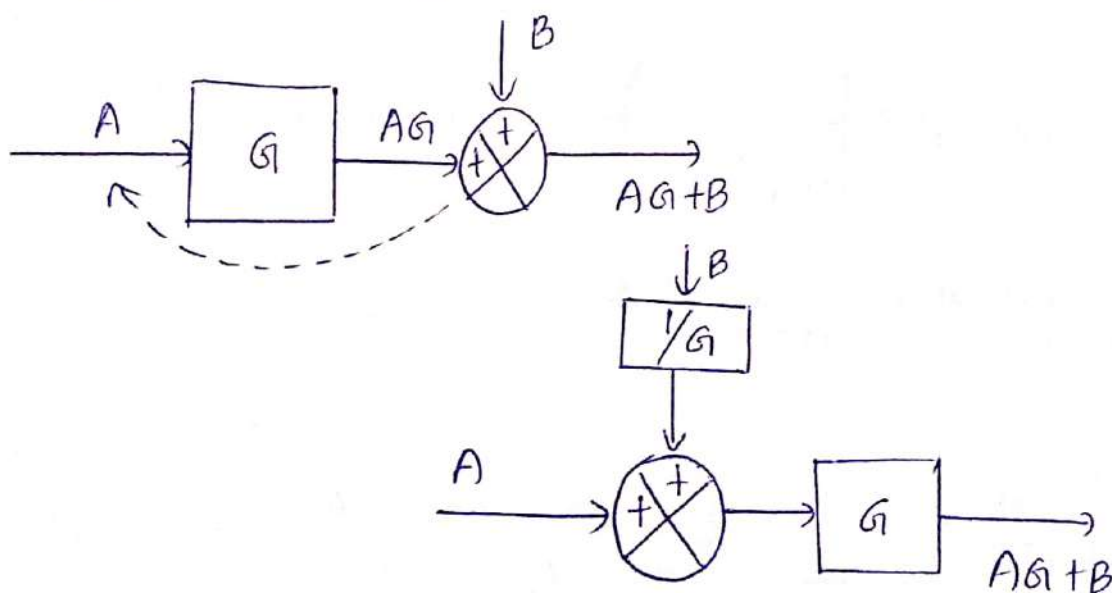
4. Moving branch point before the block



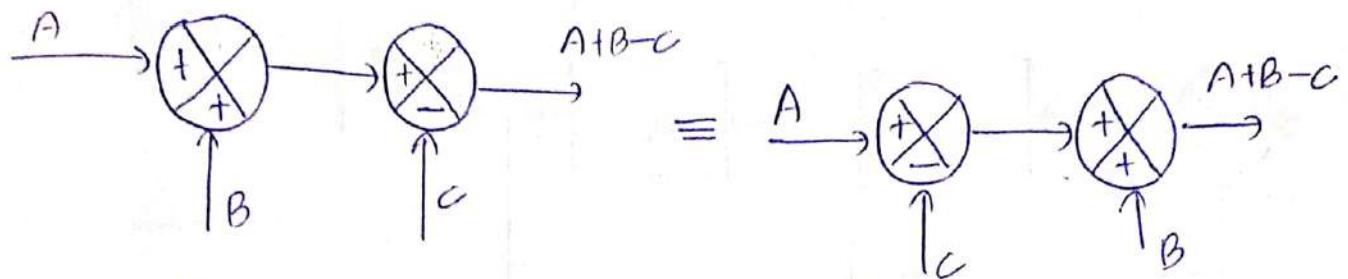
5. Moving the summing point ahead of the block



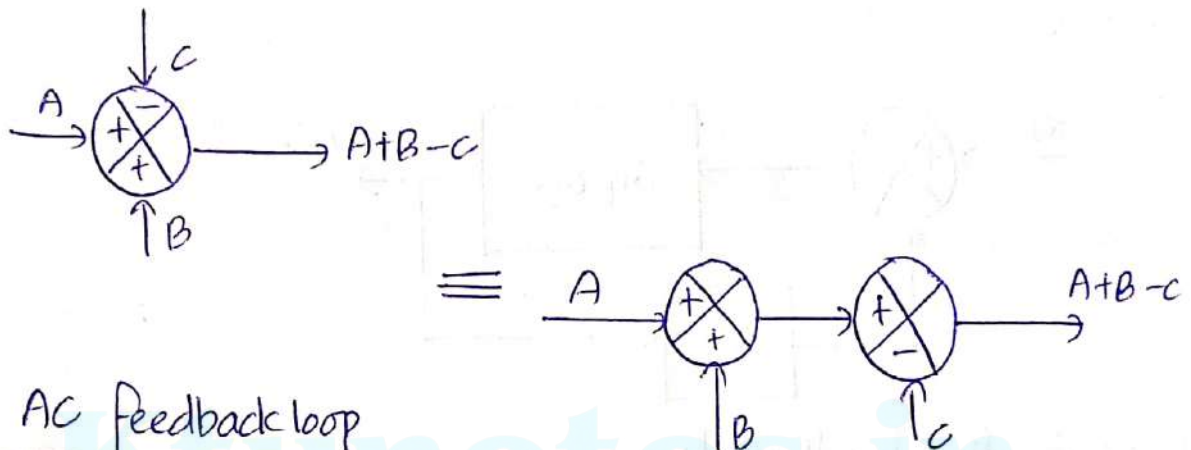
Moving the summing point before the block



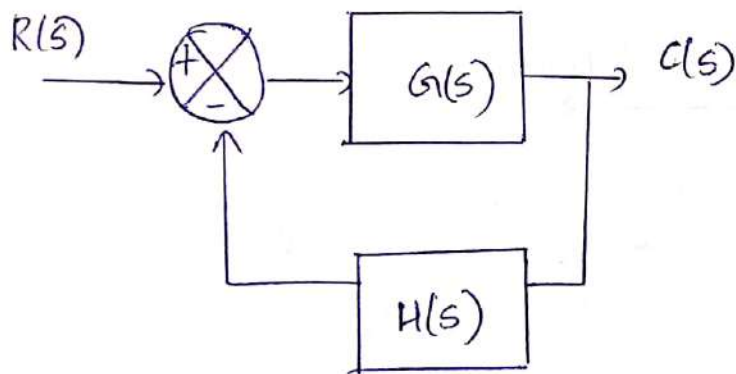
Interchanging summing points



Splitting the summing points

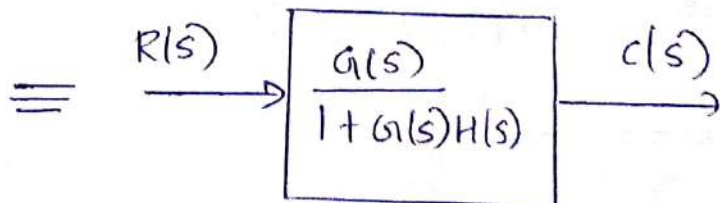


Reducing AC feedback loop



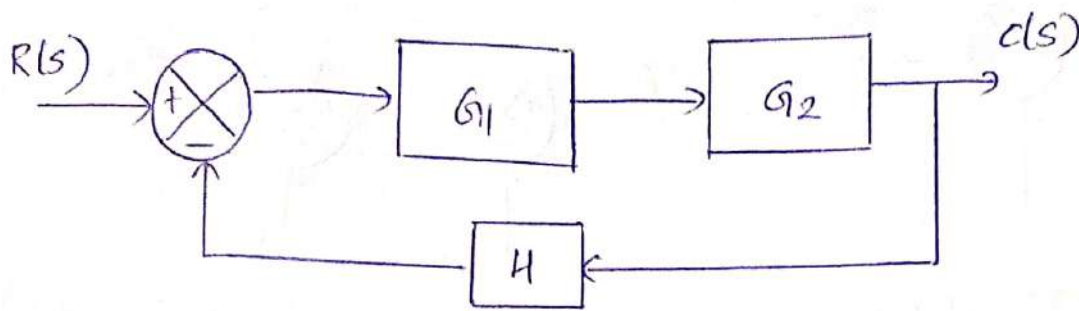
$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

negative feedback:

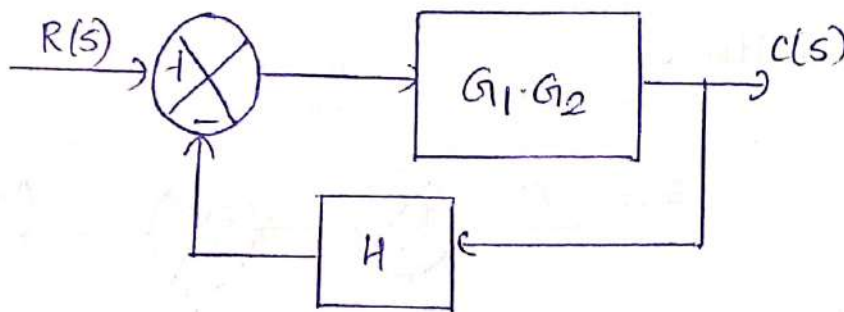


$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)}$$

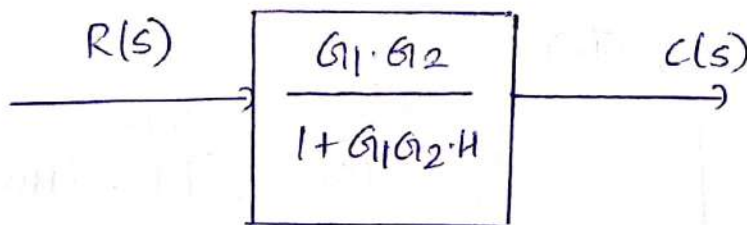
Qn: Using the block diagram reduction, find the transfer fn of the system.



1. Combining the cascade blocks G_1 and G_2 .

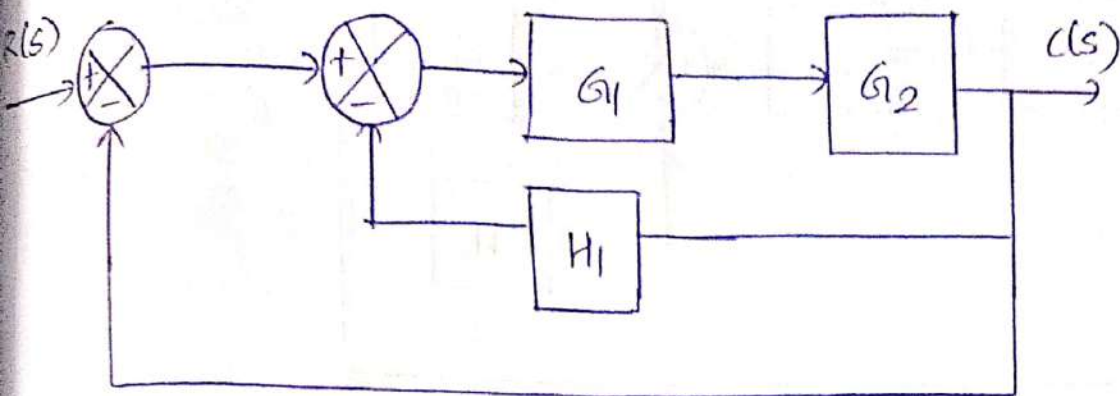


2. Eliminate the feedback loop.

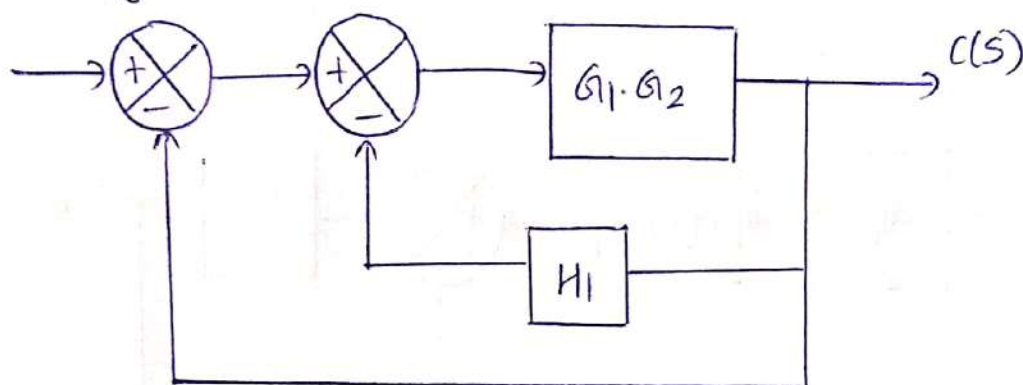


$$\frac{C(s)}{R(s)} = \frac{G_1 \cdot G_2}{1 + G_1 \cdot G_2 \cdot H}$$

Qn: Find the transfer fn using block diagram reduction.

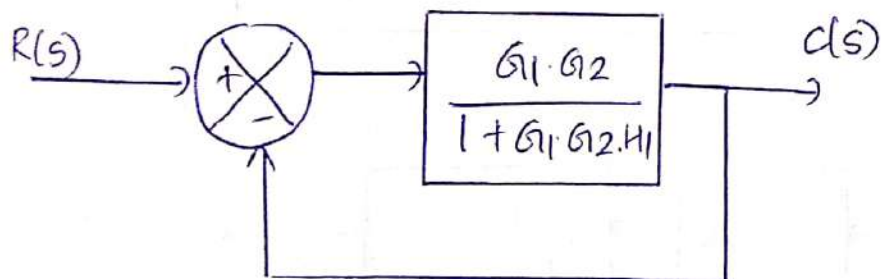


1. Combining the cascade blocks



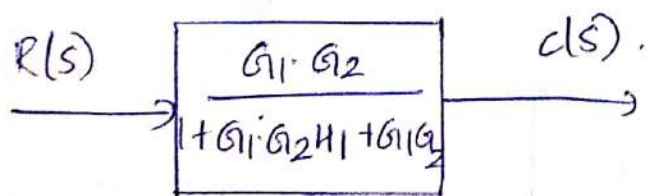
$$\frac{G}{1+GH}$$

2. Eliminating feedback loop



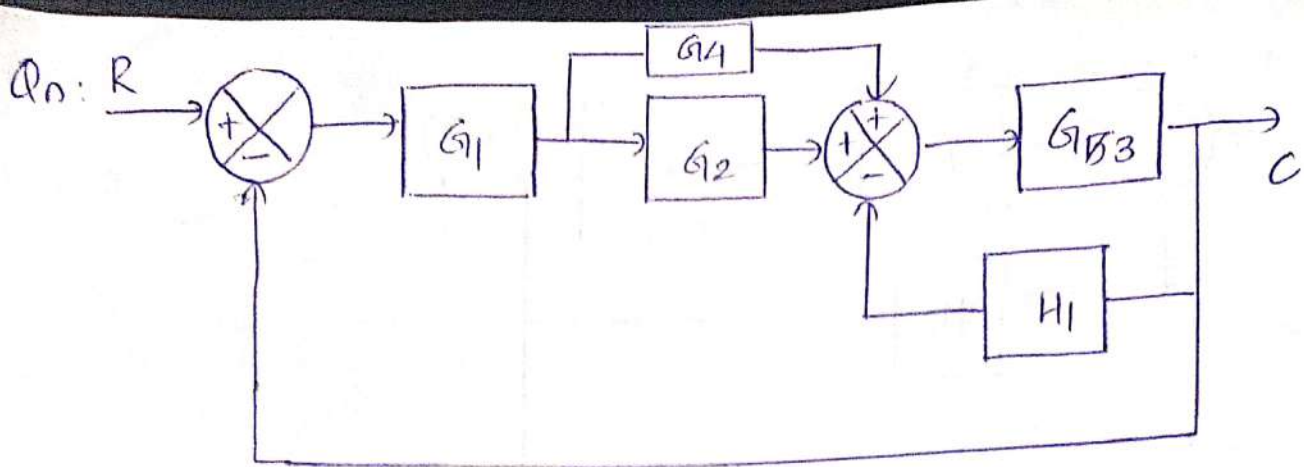
$$\frac{\cancel{G_1 G_2} \cdot G_1 G_2}{1 + G_1 G_2 H_1} \cdot \frac{1}{1 + \frac{G_1 G_2}{1 + G_1 G_2 H_1}}$$

[Unity feedback, $H=1$]

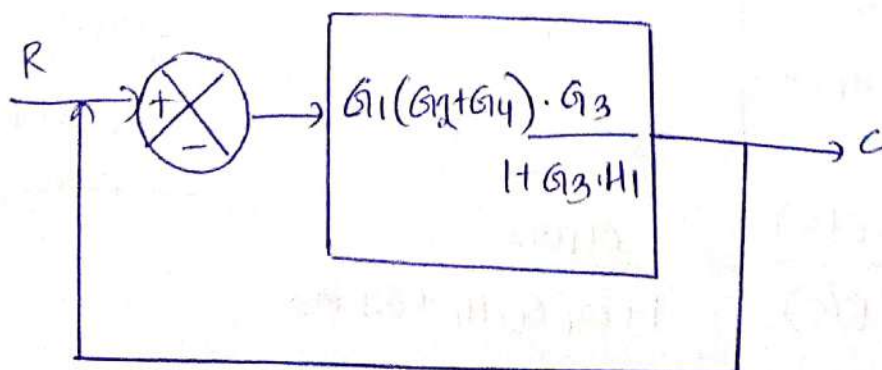
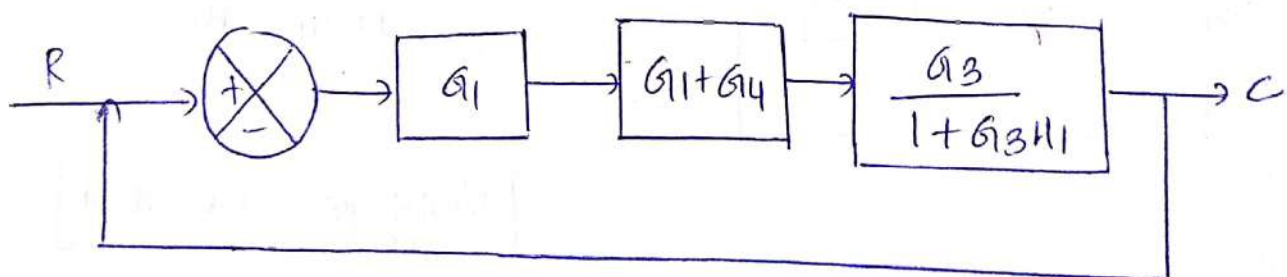
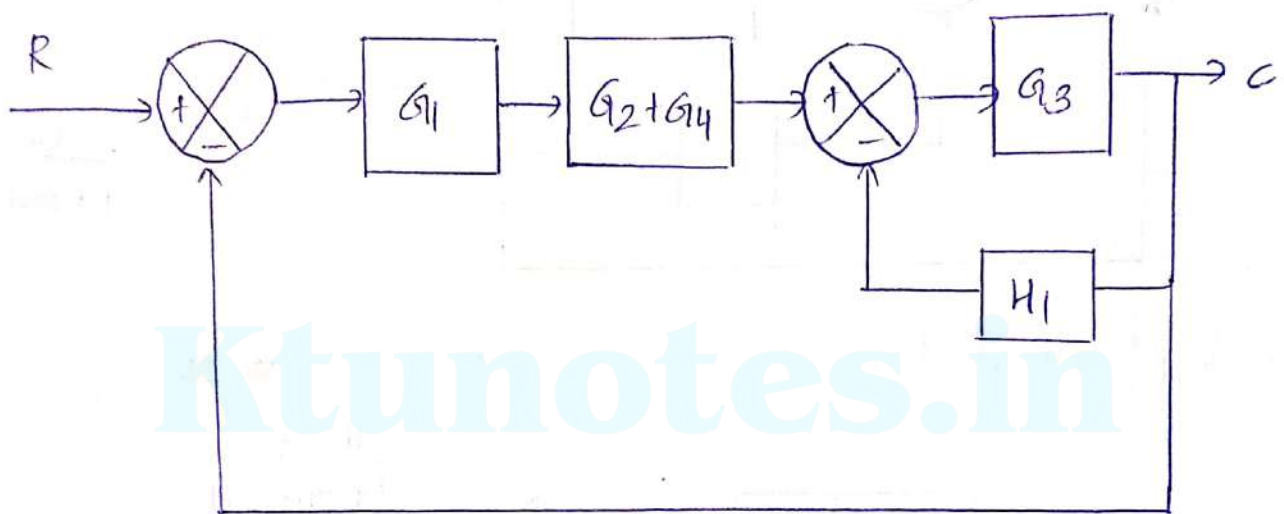


$$= \frac{G_1 G_2}{1 + G_1 G_2 H_1 + G_1 G_2}$$

$$\frac{C(s)}{R(s)} = \frac{G_1 G_2}{1 + G_1 G_2 H_1 + G_1 G_2}$$



Blocks in parallel:



$$\frac{G_1(G_2+G_4) \cdot G_3}{1 + G_3 H_1} = \frac{G_1(G_2+G_4) G_3}{1 + G_3 H_1 + G_1 G_3 (G_2+G_4)}$$

$$1 + \frac{G_1(G_2+G_4)G_3}{1 + G_3 H_1}$$

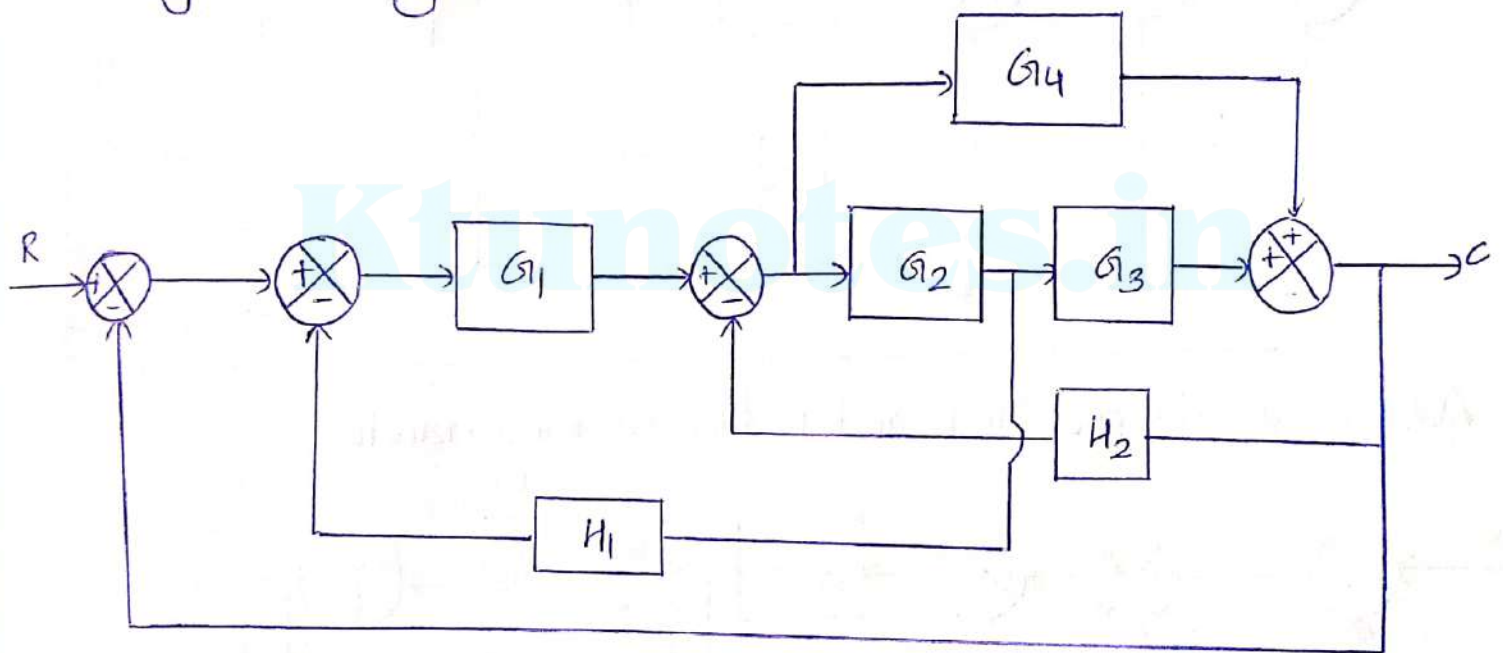
$$1 + G_3 H_1$$

$$\frac{C(s)}{R(s)}$$

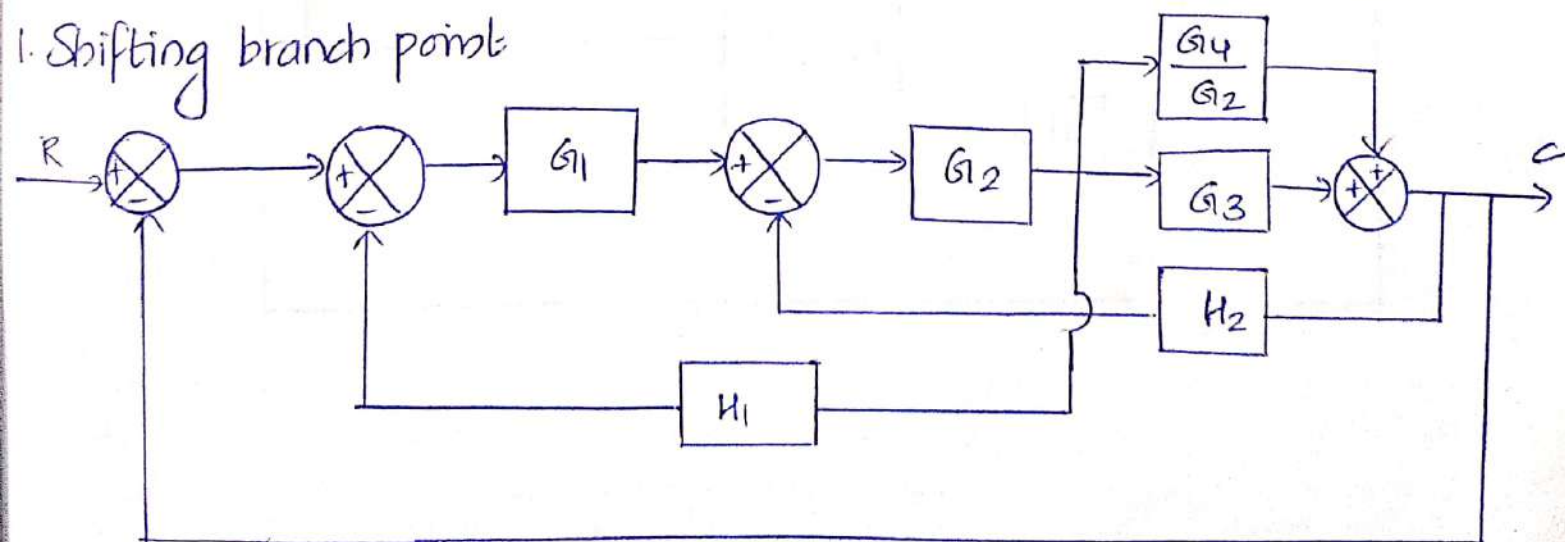
$$= \frac{G_1 G_2 G_3 + G_1 G_3 G_4}{1 + G_3 H_1 + G_1 G_2 G_3 + G_1 G_3 G_4}$$

$$1 + G_3 H_1 + G_1 G_2 G_3 + G_1 G_3 G_4$$

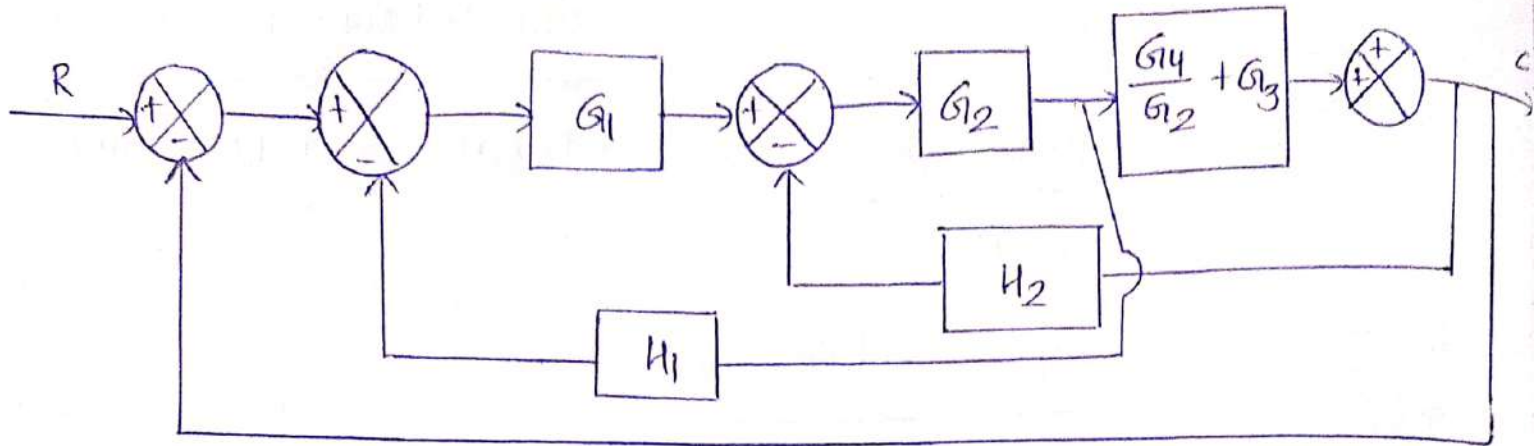
Q: Using block diagram reduction method, find the transfer function.



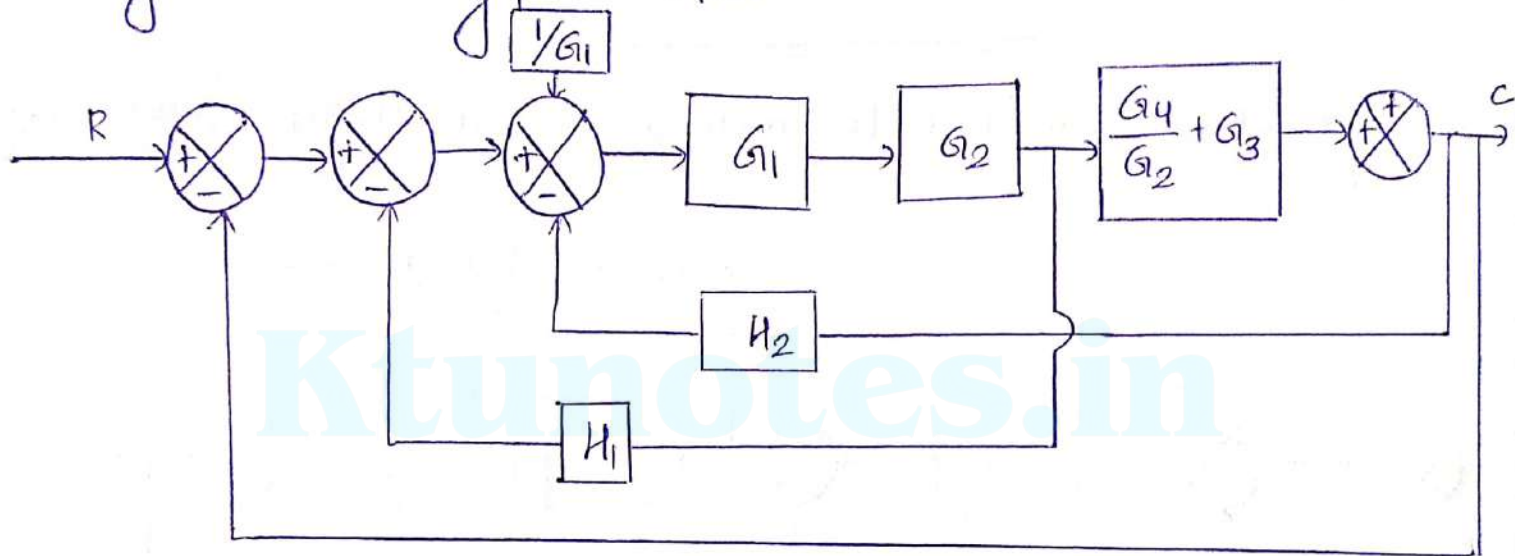
1. Shifting branch point



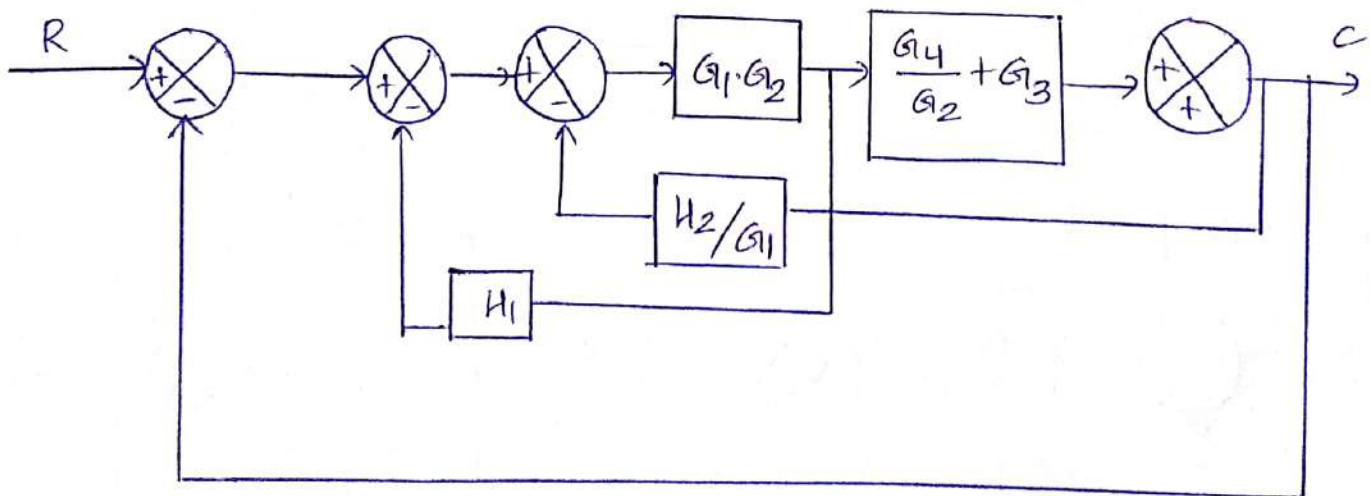
2) $G_4 \parallel G_2 \parallel G_3$ (Blocks in parallel)



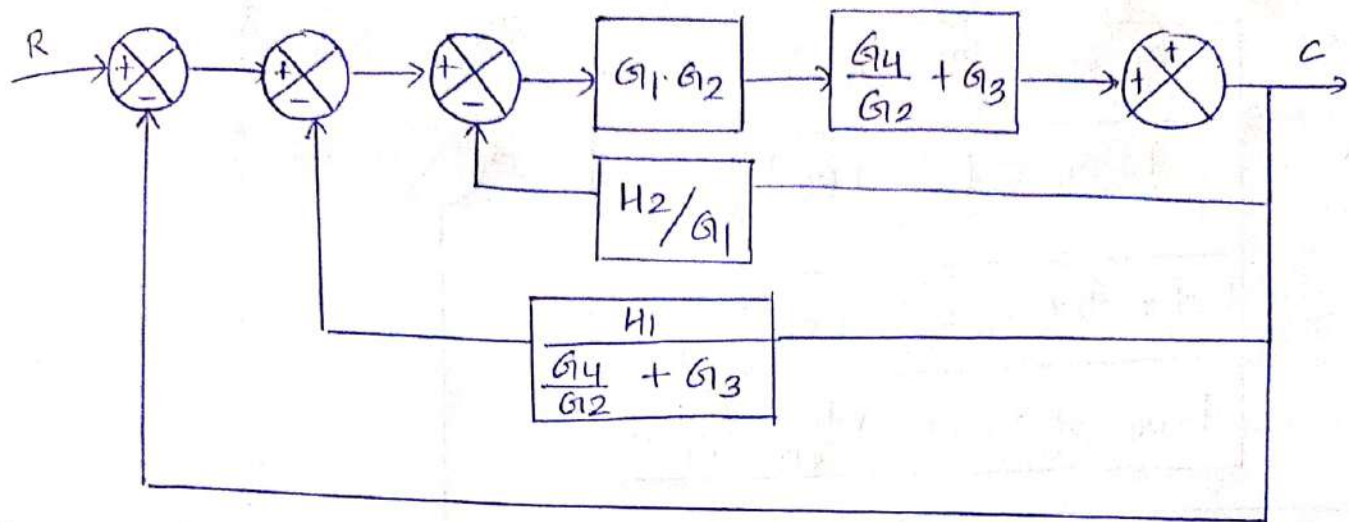
3) Moving the summing point before the block.



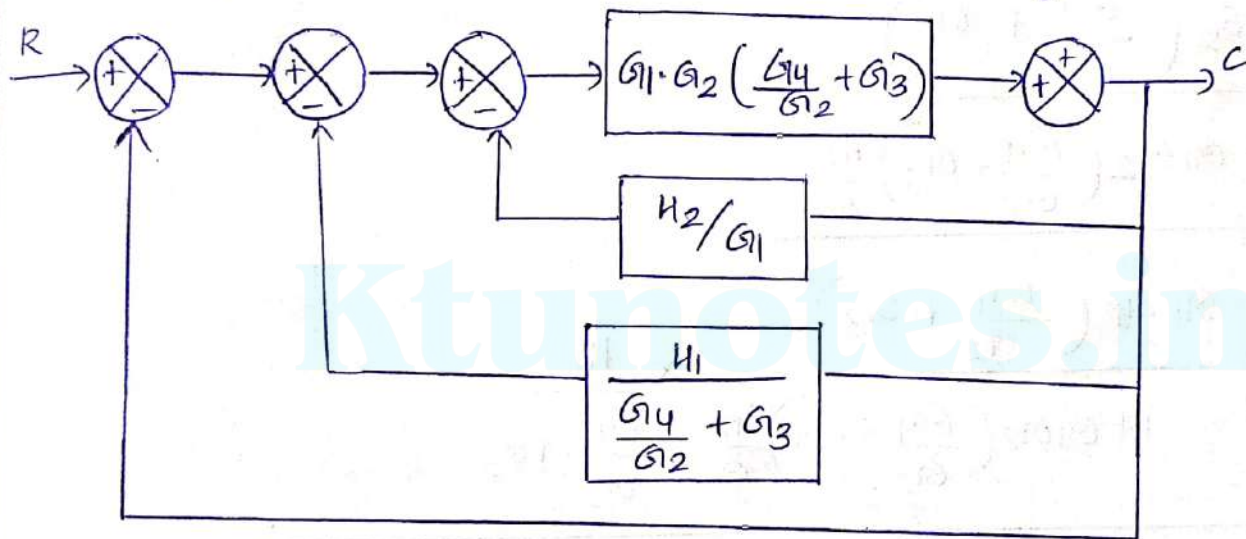
4) Add $G_1 \cdot G_2$ in one block and reduce summing branch



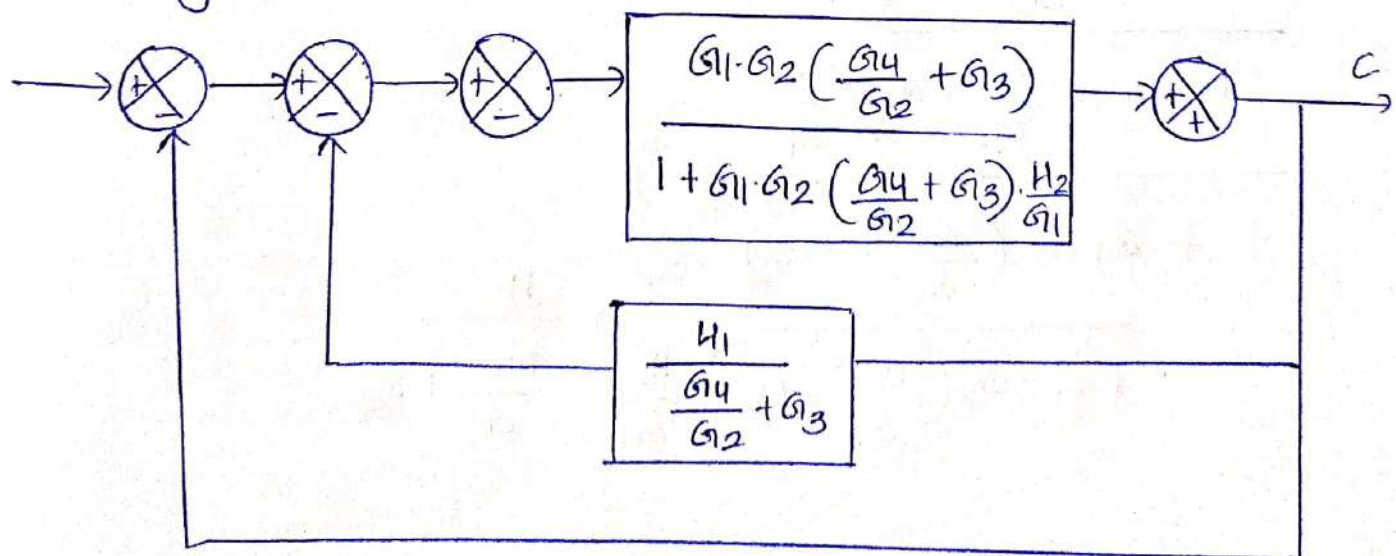
5) Shifting the branch



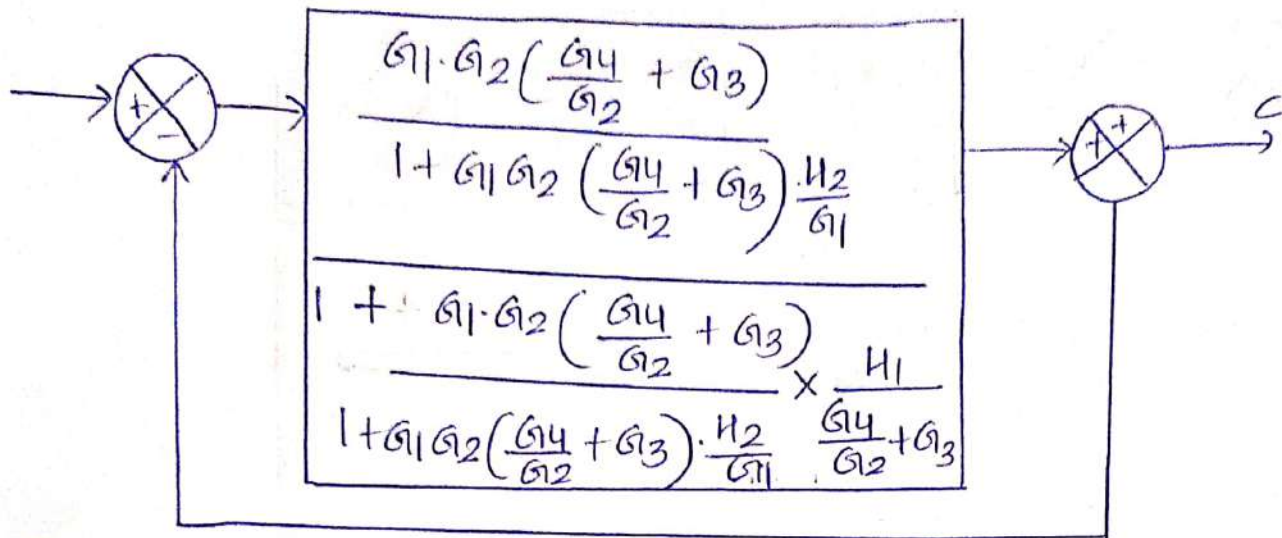
6) Reducing $G_1 \cdot G_2$ and $\frac{G_4}{G_2} + G_3$ by cascading.



7) Reducing feedback loop



8) Reducing feedback loop



9) Using feedback loop

$$\frac{C(s)}{R(s)} = \frac{G_1 \cdot G_2 \left(\frac{G_4}{G_2} + G_3 \right)}{1 + G_1 G_2 \left(\frac{G_4}{G_2} + G_3 \right) \frac{H_2}{G_1}}$$

$$\frac{1 + G_1 \cdot G_2 \left(\frac{G_4}{G_2} + G_3 \right)}{1 + G_1 G_2 \left(\frac{G_4}{G_2} + G_3 \right) \frac{H_2}{G_1}} \times \frac{H_1}{\frac{G_4}{G_2} + G_3}$$

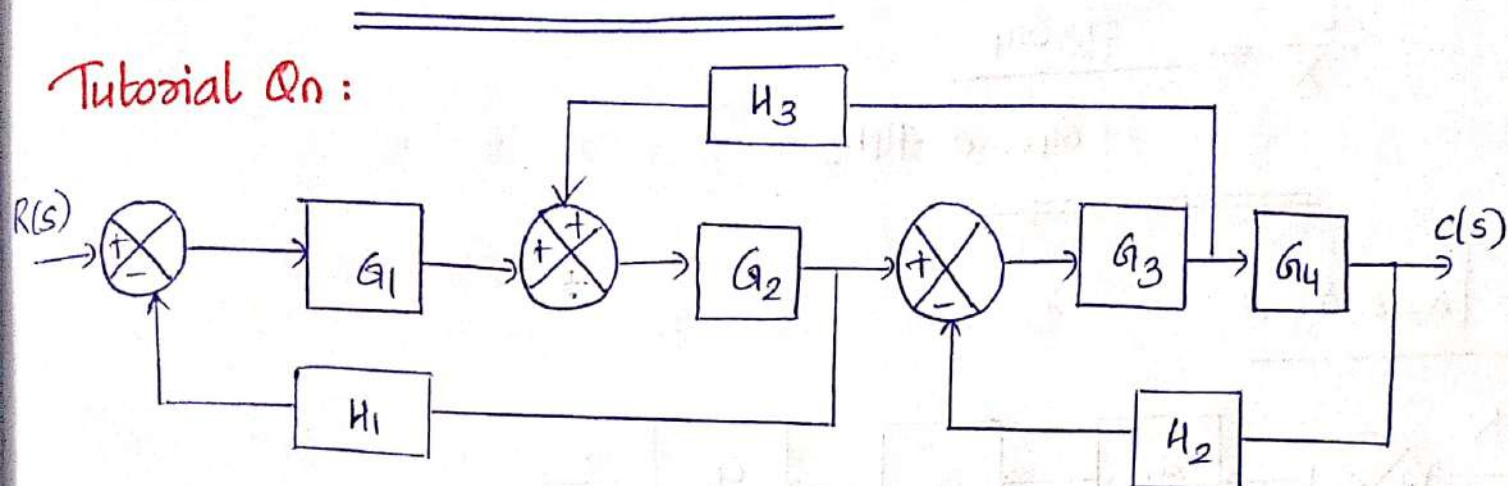
$$1 + G_1 \cdot G_2 \left(\frac{G_4}{G_2} + G_3 \right)$$

$$\frac{1 + G_1 G_2 \left(\frac{G_4}{G_2} + G_3 \right) \frac{H_2}{G_1}}{1 + G_1 G_2 \left(\frac{G_4}{G_2} + G_3 \right) \frac{H_2}{G_1}} \times \frac{H_1}{\frac{G_4}{G_2} + G_3}$$

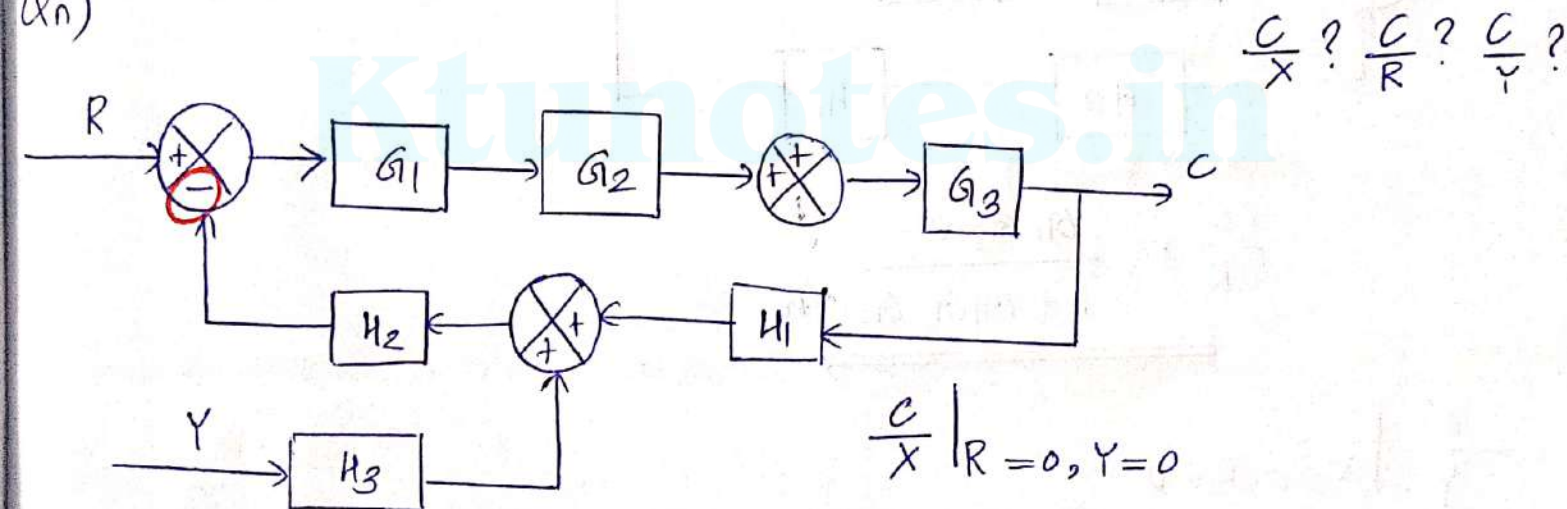
On simplification:

$$\frac{C}{R} = \frac{G_1 G_2 G_3 + G_1 G_4}{1 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_4 H_2 + G_1 G_2 G_3 + G_1 G_4}$$

Tutorial Qn:



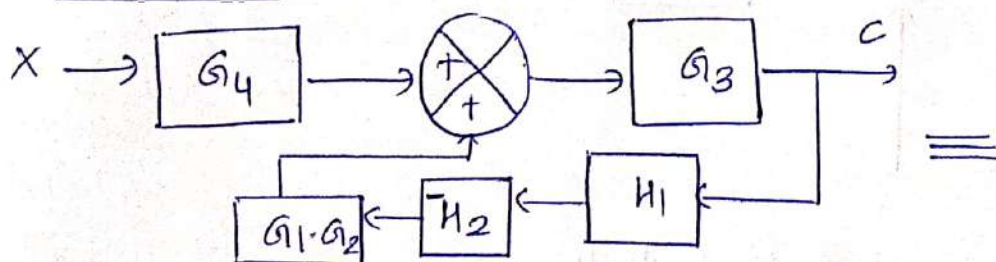
Qn)

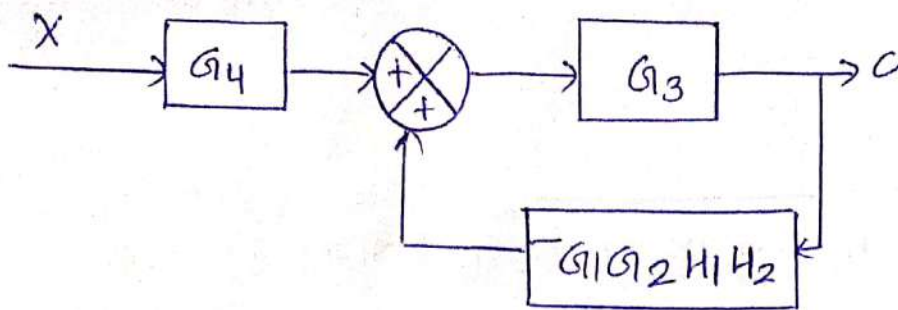


$$\frac{C}{X} \Big|_{R=0, Y=0}$$

$$\frac{C}{R} \quad \frac{C}{Y}$$

1. $R=0, Y=0$



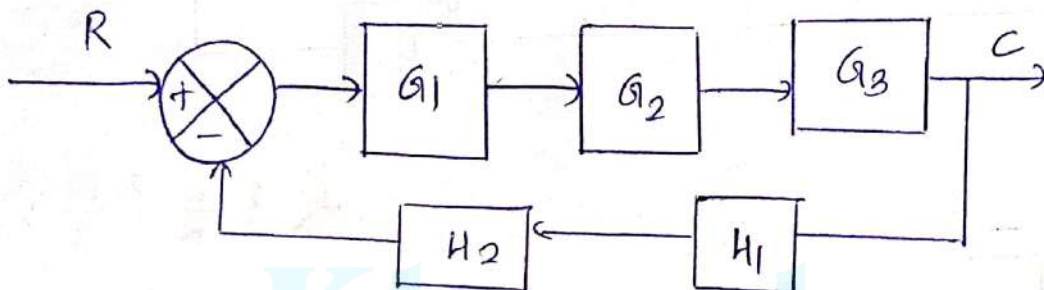


negative feedback

$$\frac{C}{X} = \frac{G_3 G_4}{1 + G_1 G_2 G_3 H_1 H_2}$$

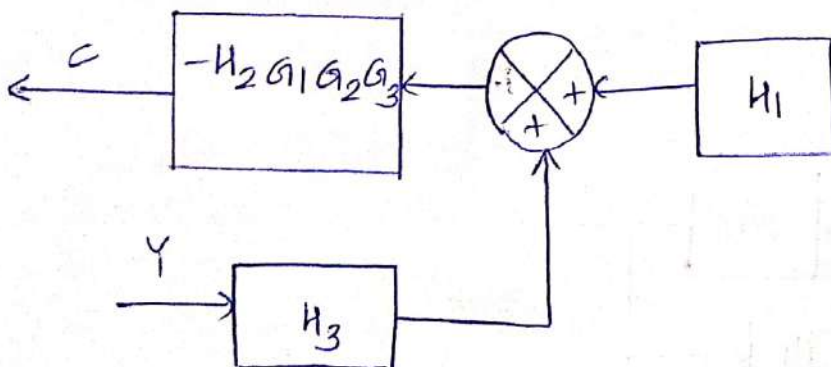
: 02 JUNE 2011

$$\frac{C}{R} \Big|_{X=0, Y=0}$$



$$\frac{C}{R} = \frac{G_1 G_2 G_3}{1 + G_1 G_2 G_3 H_1 H_2}$$

$$\frac{C}{Y} \Big|_{X=0, R=0}$$



$$\frac{C}{Y} \Big|_{X=0, R=0} = \frac{-G_1 G_2 G_3 H_2 H_3}{1 + G_1 G_2 G_3 H_1 H_2}$$

Total response :

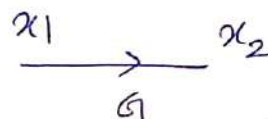
$$C = \left(\frac{G_1 G_2 G_3}{1 + G_1 G_2 G_3 H_1 H_2} \right) R + \left(\frac{-G_1 G_2 G_3 H_2 H_3}{1 + G_1 G_2 G_3 H_1 H_2} \right) Y + \underline{\underline{\left(\frac{G_3 G_4}{1 + G_1 G_2 G_3 H_1 H_2} \right) X}}$$

Signal flow graph

Represents set of linear algebraic eqns.

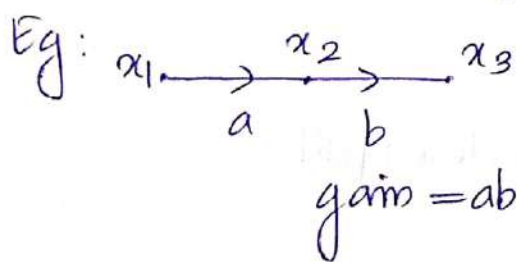


$$x_2 = G x_1$$



x_1, x_2 — nodes — signal/variable
branch

Transmittance is the gain acquired by the signal.

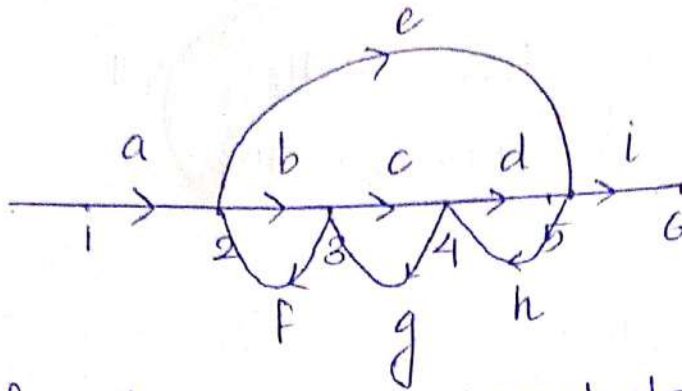


inp node : x_1 — only outgoing branches

olp node : x_2 — only incoming branches

mixed node : x_3 — both incoming and outgoing branches.

Forward path



Starts from i/p node and reach o/p node. That set of branches — forward path.

Each node should be crossed only once.

① 1-2-3-4-5-6

② 1-2-5-6

Forward path gain

① a b c d e i

② a e i

Identification of loops in the system

① 2-3-2

② 3-4-3

③ 4-5-4

④ 2-5-4-3-2

smallest closed path

Loop gain

① b f

② g g

③ d h

④ e h g f

Identification of two non-touching loops

① 2-3-2 and 4-5-4

3 non touching loops — nill (no such loops)

Mason's Gain Formula

$$TF = \frac{1}{\Delta} \sum_k P_k \Delta_k$$

k: no of forward paths

$\Delta = 1 - (\text{sum of individual loop gain}) +$
(sum of gain product of all possible combinations of two 2
non-touching loops) + (sum of gain product of all the possible
combinations of 3 non-touching loops)

Δ_k — that part of Δ which is not a part of k^{th} forward path.

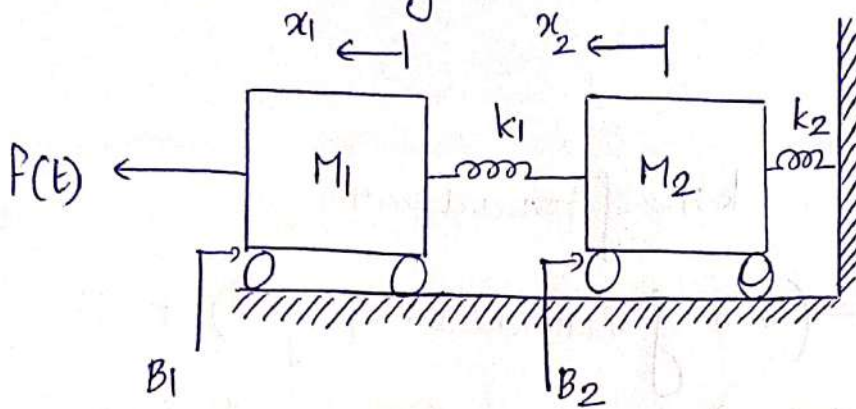
P_k — gain of a k^{th} forward path.

→ Non-touching loops will not have any nodes in common.

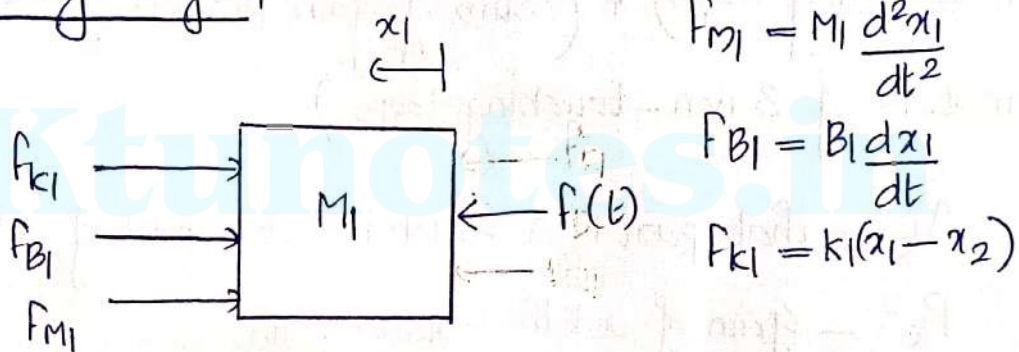
TUTORIAL

Date of submission: 05/12/2021

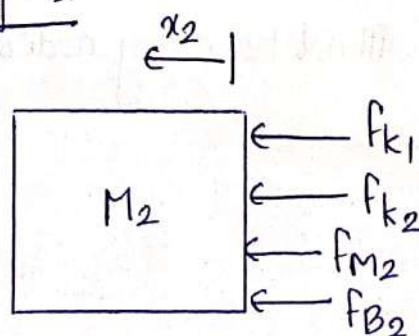
Qn: Find the transfer function $X_2(s)$ for the mechanical system. Also, draw the force-voltage and $F(s)$ force-current analogy.



Free body diagram of M_1



Free body diagram of M_2



$$F_{k1} = k_1(x_2 - x_1)$$

$$F_{k2} = k_2 x_2$$

$$F_{M2} = M_2 \frac{d^2 x_2}{dt^2}$$

$$F_{B2} = B_2 \frac{dx_2}{dt}$$

$$M_1 \frac{d^2 x_1}{dt^2} + B_1 \frac{dx_1}{dt} + k_1 (x_1 - x_2) = P(t)$$

$$M_1 s^2 X_1(s) + B_1 s X_1(s) + k_1 (X_1(s) - X_2(s)) = F(s) \text{ ————— ①}$$

$$M_2 \frac{d^2 x_2}{dt^2} + B_2 \frac{dx_2}{dt} + k_1 (x_2 - x_1) + k_2 x_2 = 0$$

$$M_2 s^2 X_2(s) + B_2 s X_2(s) + k_1 (X_2(s) - X_1(s)) + k_2 X_2(s) = 0 \text{ ————— ②}$$

$$X_2(s) [M_2 s^2 + B_2 s + k_1 + k_2] - k_1 X_1(s) = 0$$

$$k_1 X_1(s) = [M_2 s^2 + B_2 s + k_1 + k_2] X_2(s)$$

$$X_1(s) = \frac{X_2(s) (M_2 s^2 + B_2 s + k_1 + k_2)}{k_1} \text{ ————— ③}$$

Substituting ③ in ①;

$$① \Rightarrow X_1(s) [M_1 s^2 + B_1 s + k_1] - k_1 X_2(s) = F(s)$$

$$\frac{X_2(s) (M_2 s^2 + B_2 s + k_1 + k_2)}{k_1} [M_1 s^2 + B_1 s + k_1] - k_1 X_2(s) = F(s)$$

$$\frac{X_2(s) (M_2 s^2 + B_2 s + k_1 + k_2) (M_1 s^2 + B_1 s + k_1) - k_1^2 X_2(s)}{k_1} = F(s)$$

$$\frac{X_2(s)}{F(s)} = \frac{k_1}{(M_2 s^2 + B_2 s + k_1 + k_2) (M_1 s^2 + B_1 s + k_1) - k_1^2}$$

Force voltage analogy:

$$M \leftrightarrow L$$

$$B \leftrightarrow R$$

$$k \leftrightarrow \frac{1}{C}$$

$$M_1 \leftrightarrow L_1$$

$$M_2 \leftrightarrow L_2$$

$$B_1 \leftrightarrow R_1$$

and

$$B_2 \leftrightarrow R_2$$

$$K_1 \leftrightarrow \frac{1}{C_1}$$

$$K_2 \leftrightarrow \frac{1}{C_2}$$

$$\text{Force (F)} \leftrightarrow \text{voltage (V)}$$

$$\text{velocity (v}_1\text{)} \leftrightarrow \text{current (i}_1\text{)}$$

$$\text{velocity (v}_2\text{)} \leftrightarrow \text{current (i}_2\text{)}$$

$$M_1 \frac{d^2 x_1}{dt^2} + B_1 \frac{dx_1}{dt} + k_1(x_1 - x_2) = f(t) \quad \text{--- (4)}$$

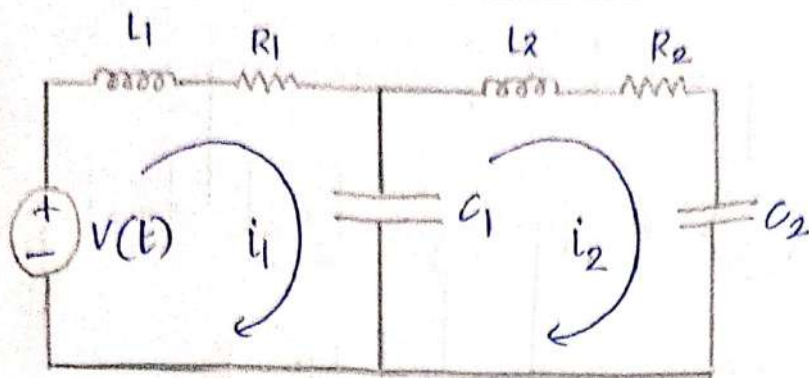
$$M_2 \frac{d^2 x_2}{dt^2} + B_2 \frac{dx_2}{dt} + k_1(x_2 - x_1) + k_2 x_2 = 0 \quad \text{--- (5)}$$

$$\text{(4)} \Rightarrow M_1 \frac{dv_1}{dt} + B_1 v_1 + k_1 \int (v_1 - v_2) dt = V(t) \quad \text{--- (6)}$$

$$\text{(5)} \Rightarrow M_2 \frac{dv_2}{dt} + B_2 v_2 + k_1 \int (v_2 - v_1) dt + k_2 \int v_2 dt = 0 \quad \text{--- (7)}$$

$$L_1 \frac{di_1}{dt} + R_1 i_1 + \frac{1}{C_1} \int (i_1 - i_2) dt = V(t) \quad \text{--- (I)}$$

$$L_2 \frac{di_2}{dt} + R_2 i_2 + \frac{1}{C_1} \int (i_2 - i_1) dt + \frac{1}{C_2} \int i_2 dt = 0 \quad \text{--- (II)}$$



Force-current analogy:

$$M \leftrightarrow C$$

$$B \leftrightarrow \frac{1}{R}$$

$$k \leftrightarrow \frac{1}{L}$$

$$M_1 \leftrightarrow C_1$$

$$B_1 \leftrightarrow \frac{1}{R_1}$$

$$k_1 \leftrightarrow \frac{1}{L_1}$$

$$M_2 \leftrightarrow C_2$$

$$B_2 \leftrightarrow \frac{1}{R_2}$$

$$k_2 \leftrightarrow \frac{1}{L_2}$$

$$F(t) \leftrightarrow i(t)$$

$$\text{velocity } (v_1) \leftrightarrow \text{voltage } (V_1)$$

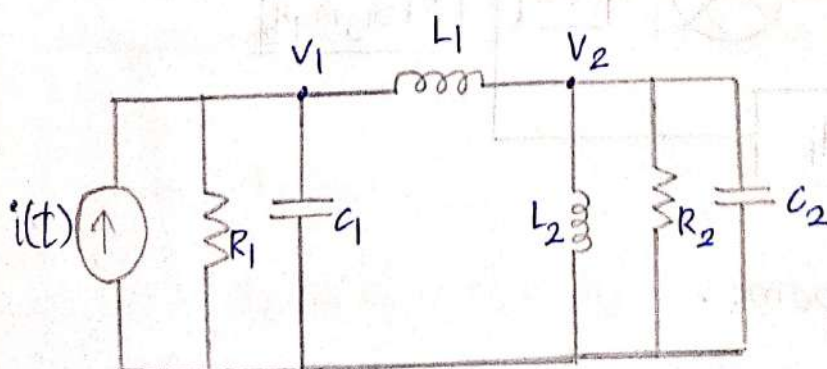
$$\text{velocity } (v_2) \leftrightarrow \text{Voltage } (V_2)$$

$$\textcircled{6} \Rightarrow M_1 \frac{dv_1}{dt} + B_1 v_1 + k_1 \int (v_1 - v_2) dt = i(t)$$

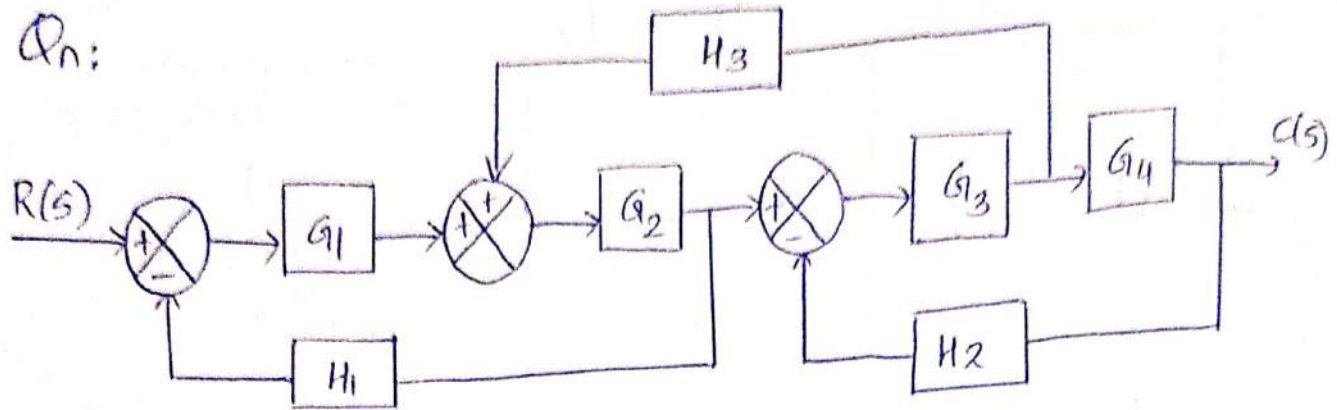
$$\textcircled{7} \Rightarrow M_2 \frac{dv_2}{dt} + B_2 v_2 + k_1 \int (v_2 - v_1) dt + k_2 \int v_2 dt = 0$$

$$C_1 \frac{dV_1}{dt} + \frac{1}{R_1} V_1 + \frac{1}{L_1} \int (V_1 - V_2) dt = i(t) \quad \text{--- (III)}$$

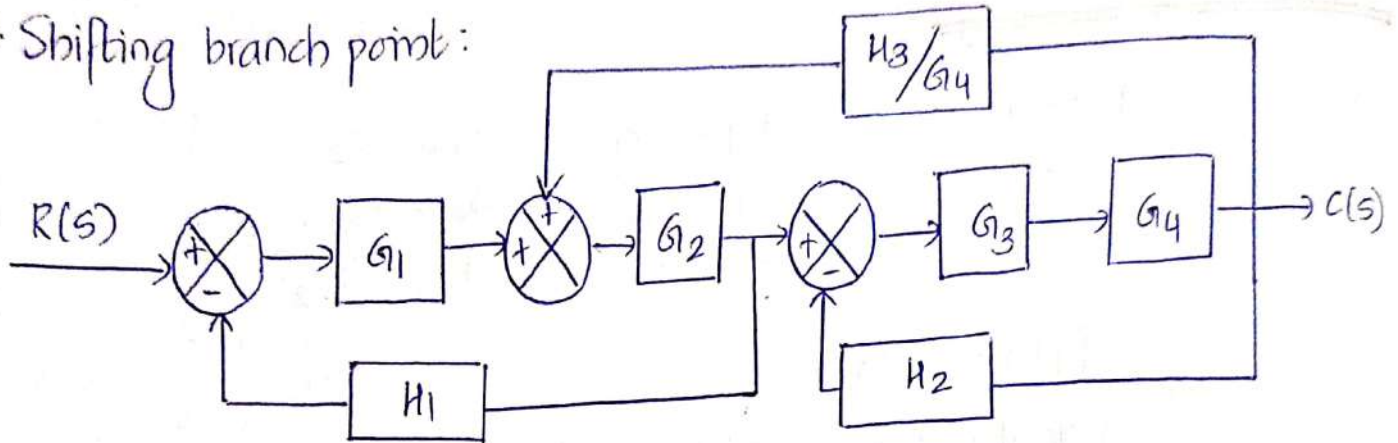
$$C_2 \frac{dV_2}{dt} + \frac{1}{R_2} V_2 + \frac{1}{L_1} \int (V_2 - V_1) dt + \frac{1}{L_2} \int V_2 dt = 0 \quad \text{--- (IV)}$$



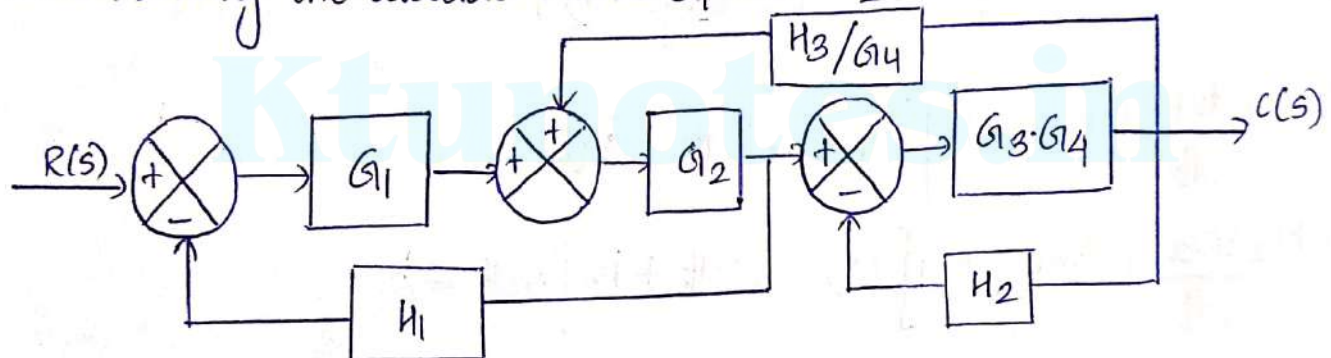
Qn:



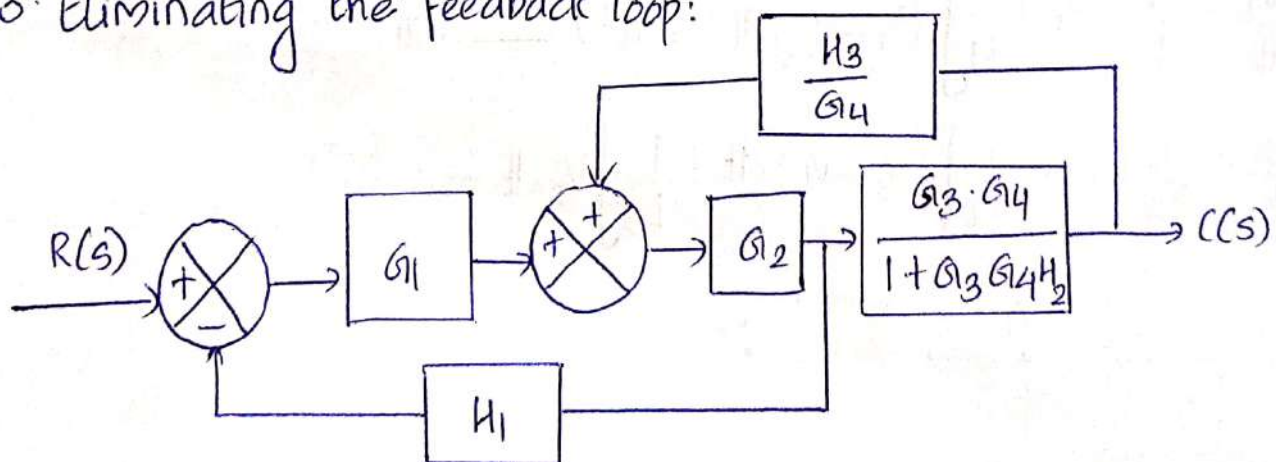
1. Shifting branch point:



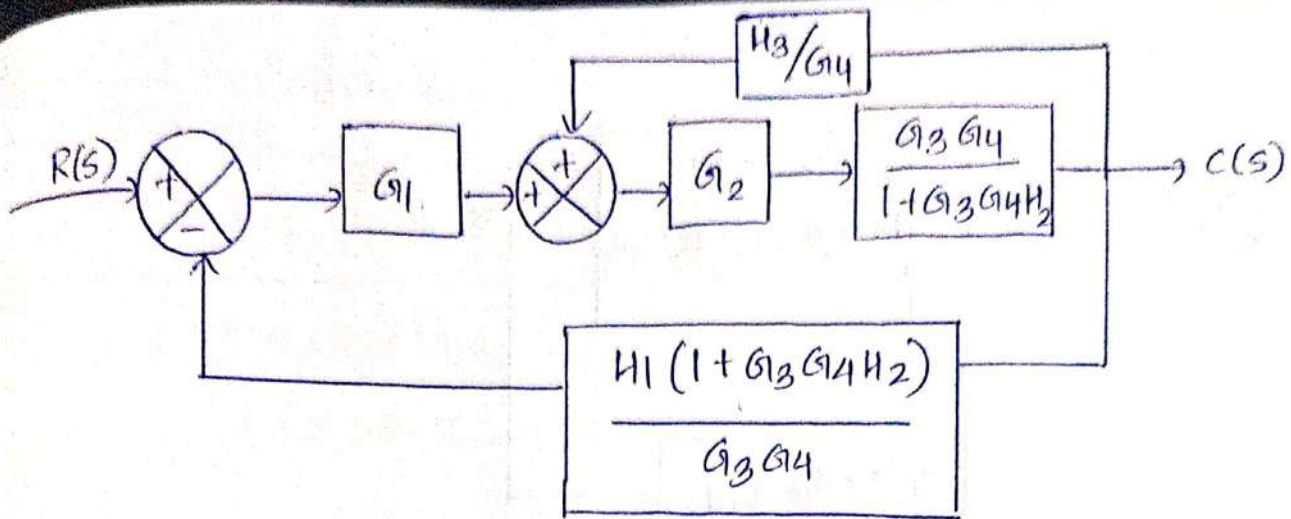
2. Combining the cascade blocks G_1 and G_2 :



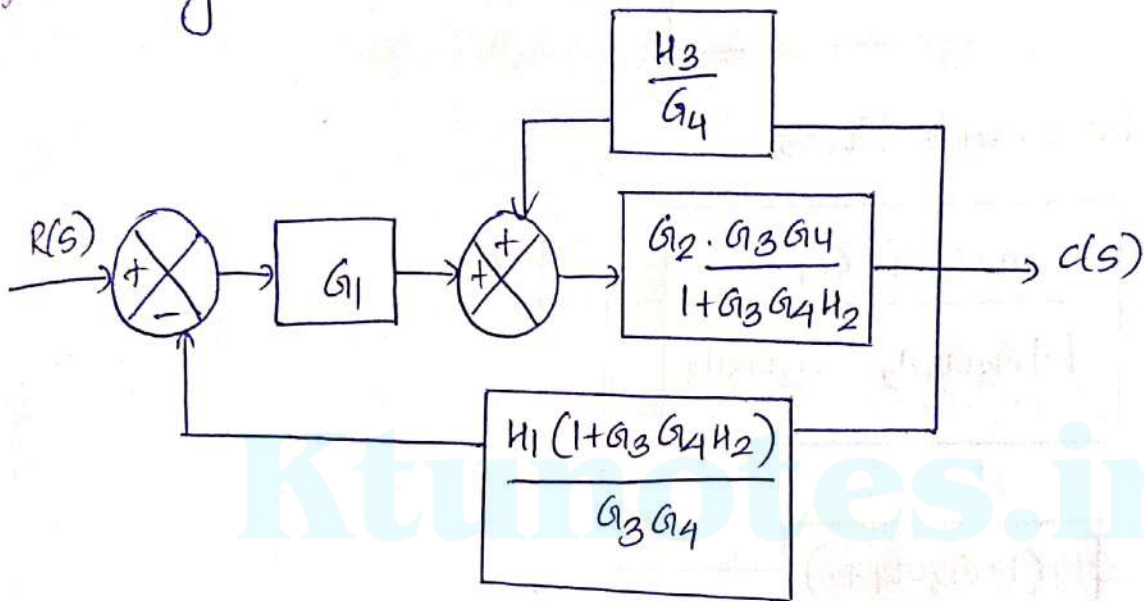
3. Eliminating the feedback loop:



4. Shifting the branch point:

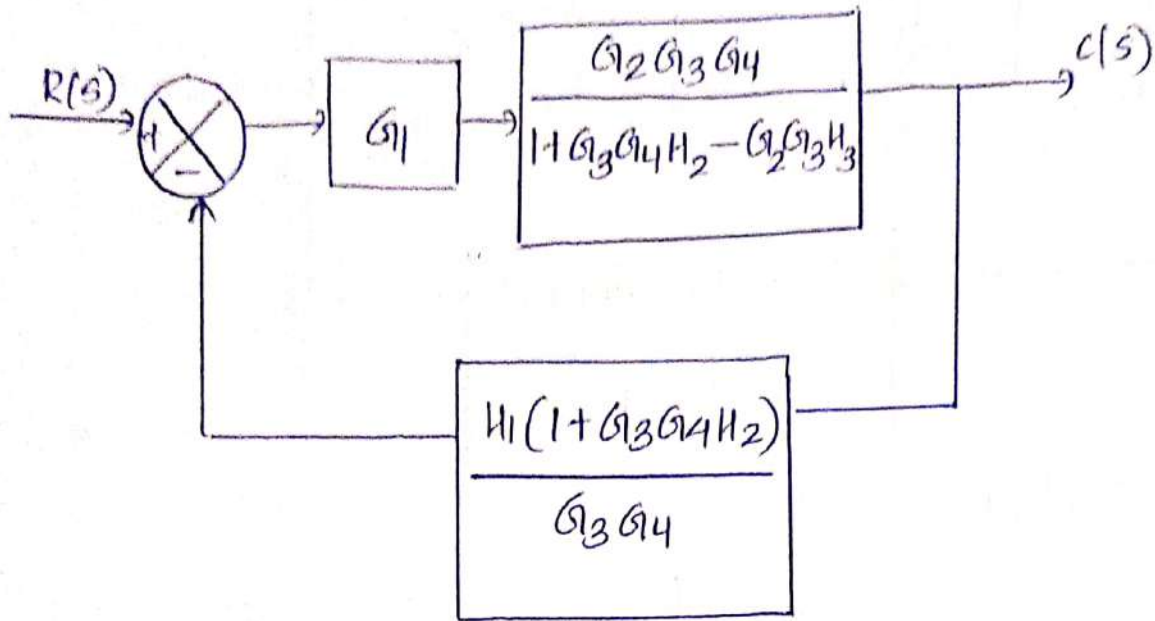


5. Combining the cascade blocks:

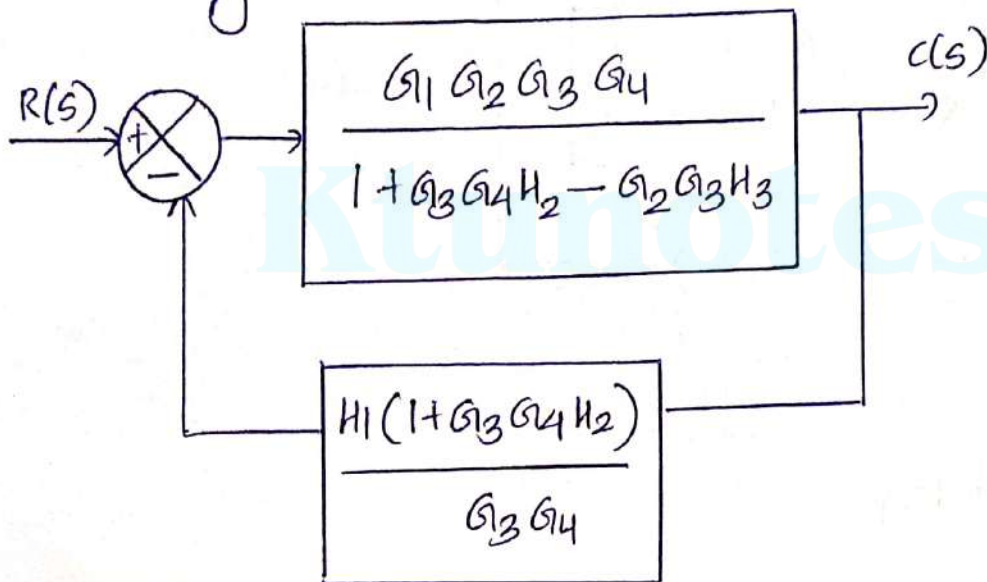


6. Eliminating the feedback loop:

$$\begin{aligned}
 & \frac{\frac{G_2 G_3 G_4}{1 + G_3 G_4 H_2}}{1 - \frac{G_2 G_3 G_4}{1 + G_3 G_4 H_2} \times \frac{H_3}{G_4}} = \frac{\frac{G_2 G_3 G_4}{1 + G_3 G_4 H_2}}{1 - \frac{G_2 G_3 H_3}{1 + G_3 G_4 H_2}} \\
 & = \frac{G_2 G_3 G_4}{1 + G_3 G_4 H_2 - G_2 G_3 H_3}
 \end{aligned}$$



7. Combining the cascade blocks:



8. Eliminating the feedback loop:

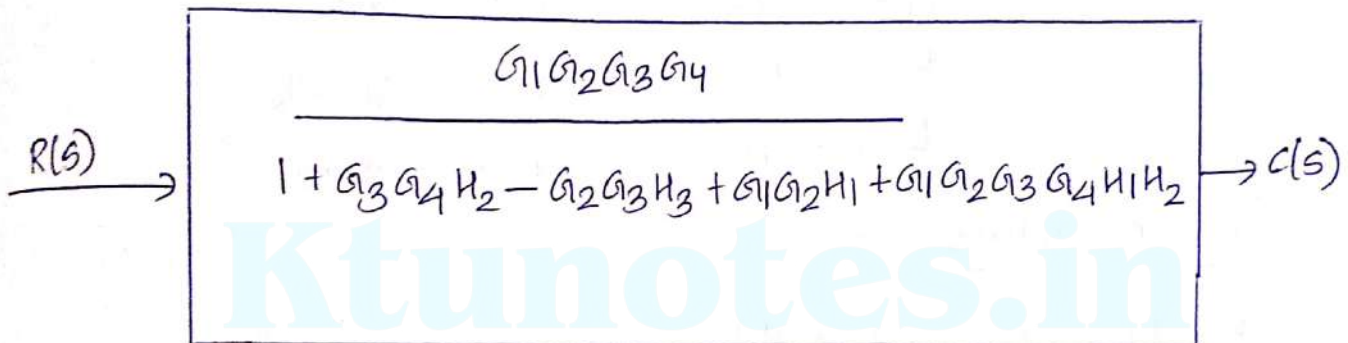
$$\frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2 - G_2 G_3 H_3}$$

$$1 + \left(\frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2 - G_2 G_3 H_3} \times \frac{H_1 (1 + G_3 G_4 H_2)}{G_3 G_4} \right)$$

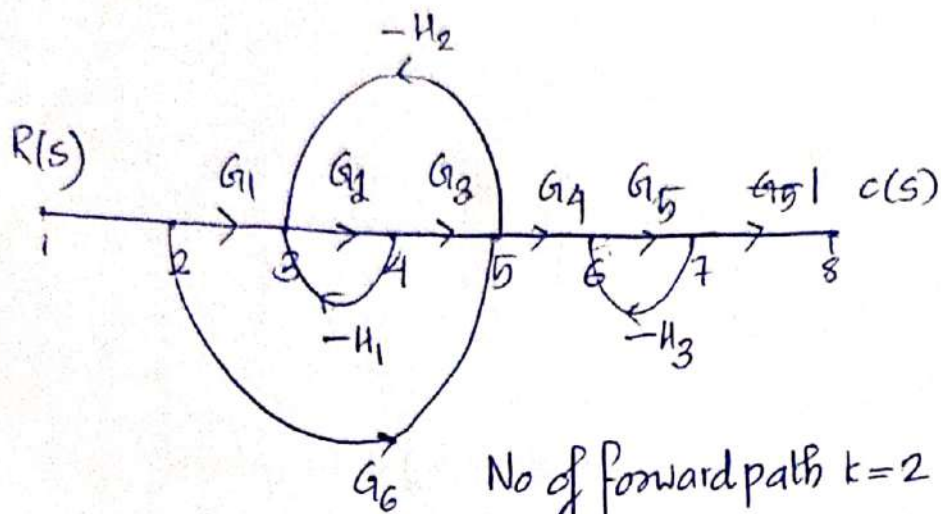
$$= \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2 - G_2 G_3 H_3}$$

$$= \frac{1 + G_1 G_2 H_1 (1 + G_3 G_4 H_2)}{1 + G_3 G_4 H_2 - G_2 G_3 H_3}$$

$$= \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2 - G_2 G_3 H_3 + G_1 G_2 H_1 + G_1 G_2 H_1 G_3 G_4 H_2}$$



$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2 - G_2 G_3 H_3 + G_1 G_2 H_1 + G_1 G_2 H_1 G_3 G_4 H_2}$$



No of forward path $k=2$
 Forward path gain $= P_1 = G_1 G_2 G_3 G_4 G_5$
 $P_2 = G_4 G_5 G_6$

Individual loops:

$L_1 \rightarrow 3-4-3$	$P_{11} = -G_2 H_1$
$L_2 \rightarrow 3-4-5-3$	$P_{21} = -G_2 G_3 H_2$
$L_3 \rightarrow 6-7-6$	$P_{31} = -G_5 H_3$

Two non-touching loops

L_1 and L_3 $P_{12} = G_2 G_5 H_1 H_3$

L_2 and L_3 $P_{22} = G_2 G_3 G_5 H_2 H_3$

$$\Delta = 1 - [-G_2 H_1 - G_2 G_3 H_2 - G_5 H_3] + G_2 G_5 H_1 H_3 + G_2 G_3 G_5 H_2 H_3$$

$$= 1 + G_2 H_1 + G_2 G_3 H_2 + G_5 H_3 + G_2 G_5 H_1 H_3 + G_2 G_3 G_5 H_2 H_3$$

$$\Delta_1 = 1$$

$$\Delta_2 = G_2 H_1 + 1$$

$$= 1 + G_2 H_1$$

$$TF = \frac{1}{\Delta} \sum_k P_k \Delta_k$$

$$= \frac{1}{\Delta} [P_1 \Delta_1 + P_2 \Delta_2]$$

$$= \frac{G_1 G_2 G_3 G_4 G_5 + G_4 G_5 G_6 (1 + G_2 H_1)}{1 + G_2 H_1 + G_2 G_3 H_2 + G_5 H_3 + G_2 G_5 H_1 H_3 + G_2 G_3 G_5 H_2 H_3}$$

$$= \frac{G_1 G_2 G_3 G_4 G_5 + G_4 G_5 G_6 + G_2 G_4 G_5 G_6 H_1}{1 + G_2 H_1 + G_2 G_3 H_2 + G_5 H_3 + G_2 G_5 H_1 H_3 + G_2 G_3 G_5 H_2 H_3}$$

Qn: For the following linear eqns, find out the transfer P_n :

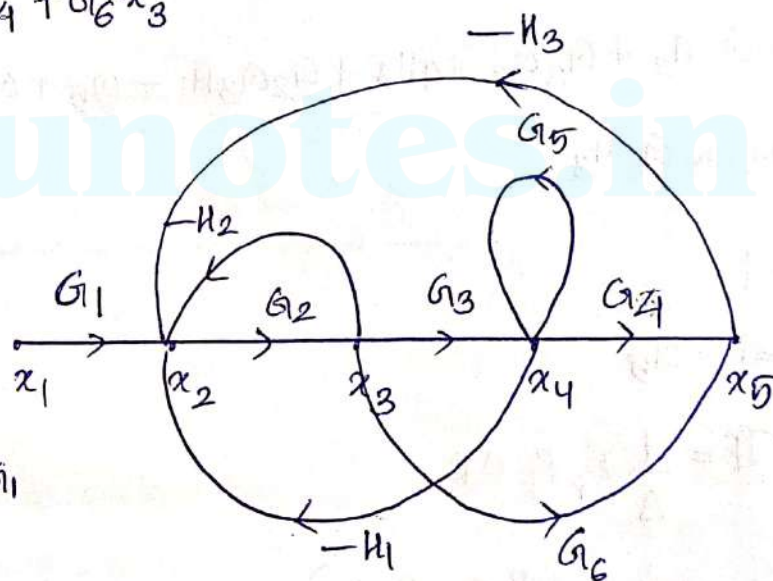
$$x_2 = \overset{G_1}{x_1} - H_2 x_3 - H_1 x_4 - H_3 x_5$$

$$x_3 = G_2 x_2$$

$$x_4 = G_3 x_3 + G_5 x_5$$

$$x_5 = G_4 x_4 + G_6 x_3$$

Find $\frac{x_5(s)}{x_1(s)}$.



No. of fwd path = 2

$$P_1 = G_2 G_3 G_4 G_1$$

$$P_2 = G_1 G_2 G_6$$

$$L_1, x_2 - x_2 - x_3 - x_2$$

$$P_{11} = -G_2 H_2$$

$$L_2, x_2 - x_3 - x_4 - x_5 - x_2$$

$$P_{21} = -G_2 G_3 G_4 H_3$$

$$L_3, x_2 - x_3 - x_4 - x_2$$

$$P_{31} = -G_2 G_3 H_1$$

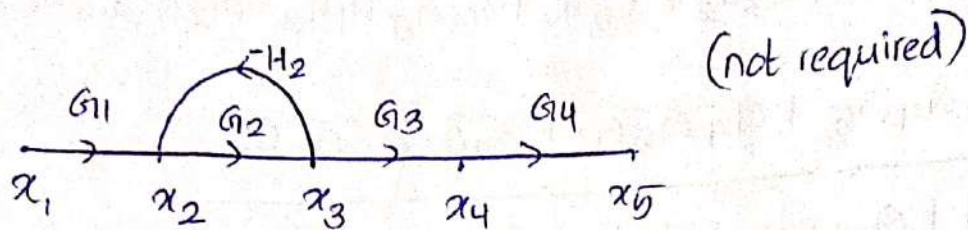
$$L_4, x_3 - x_4 - x_5 - x_4$$

$$P_{41} = G_5$$

$$L_5, x_2 - x_3 - x_4 - x_2$$

$$L_6, x_2 - x_3 - x_5 - x_2$$

$$P_{61} = -G_2 G_6 H_3$$



Two non-touching loops

$$L_1 \text{ and } L_4 \quad P_{12} = -G_2 G_5 H_2$$

$$L_4 \text{ and } L_5 \quad P_{22} = -G_2 G_5 G_6 H_3$$

$$\Delta = 1 - \left[-G_2 H_2 - G_2 G_3 G_4 H_3 - G_2 G_3 H_1 + G_5 - G_2 G_6 H_3 \right] +$$

$$-G_2 G_5 H_2 - G_2 G_5 G_6 H_3$$

$$= 1 + G_2 H_2 + G_2 G_3 G_4 H_3 + G_2 G_3 H_1 - G_5 + G_2 G_6 H_3 - G_2 G_5 H_2 -$$

$$G_2 G_5 G_6 H_3$$

$$\Delta_1 = 1$$

$$\Delta_2 = 1 - G_5$$

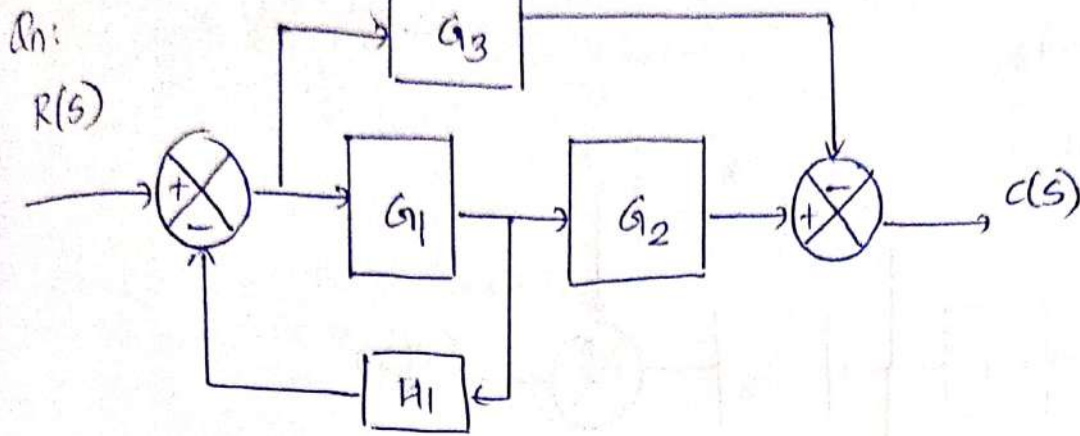
$$TF = \frac{1}{\Delta} \sum_k P_k \Delta_k$$

$$= \frac{1}{\Delta} (P_1 \Delta_1 + P_2 \Delta_2)$$

$$= \frac{G_1 G_2 G_3 G_4 + G_1 G_2 G_6 (1 - G_5)}{1 + G_2 H_2 + G_2 G_3 G_4 H_3 + G_2 G_3 H_1 - G_5 + G_2 G_6 H_3 - G_2 G_5 H_2 - G_2 G_5 G_6 H_3}$$

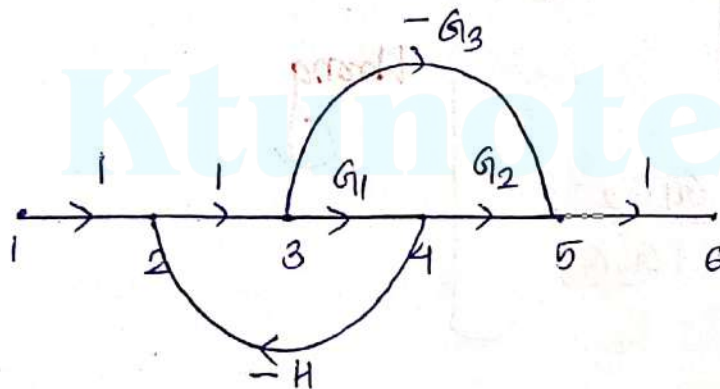
$$= \frac{G_1 G_2 G_3 G_4 + G_1 G_2 G_6 - G_1 G_2 G_5 G_6}{1 + G_2 H_2 + G_2 G_3 G_4 H_3 + G_2 G_3 H_1 - G_5 + G_2 G_6 H_3 - G_2 G_5 H_2 - G_2 G_5 G_6 H_3}$$

$$1 + G_2 H_2 + G_2 G_3 G_4 H_3 + G_2 G_3 H_1 - G_5 + G_2 G_6 H_3 - G_2 G_5 H_2 - G_2 G_5 G_6 H_3$$



Convert the block diagram into signal-flow graph and obtain the transfer fn $\frac{C(s)}{R(s)}$ and verify the ans using block diagram reduction method

💡 In order to convert block diagram to signal flow graph, assign nodes for i/p point, branch point, summing point and o/p point.



first path : 1-2-3-4-5-6

2nd path: 1-2-3-5-6

$$P_1 = G_1 G_2$$

$$P_2 = -G_3$$

$$L_1, 2, 3-4-2 \quad P_{11} = -G_1 H$$

$t_{2,3}$

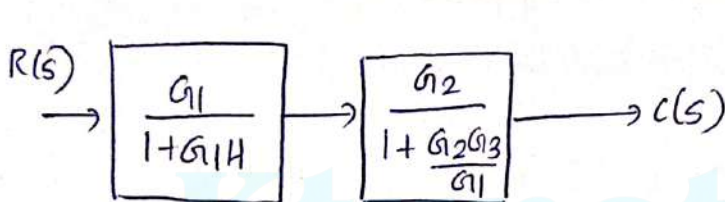
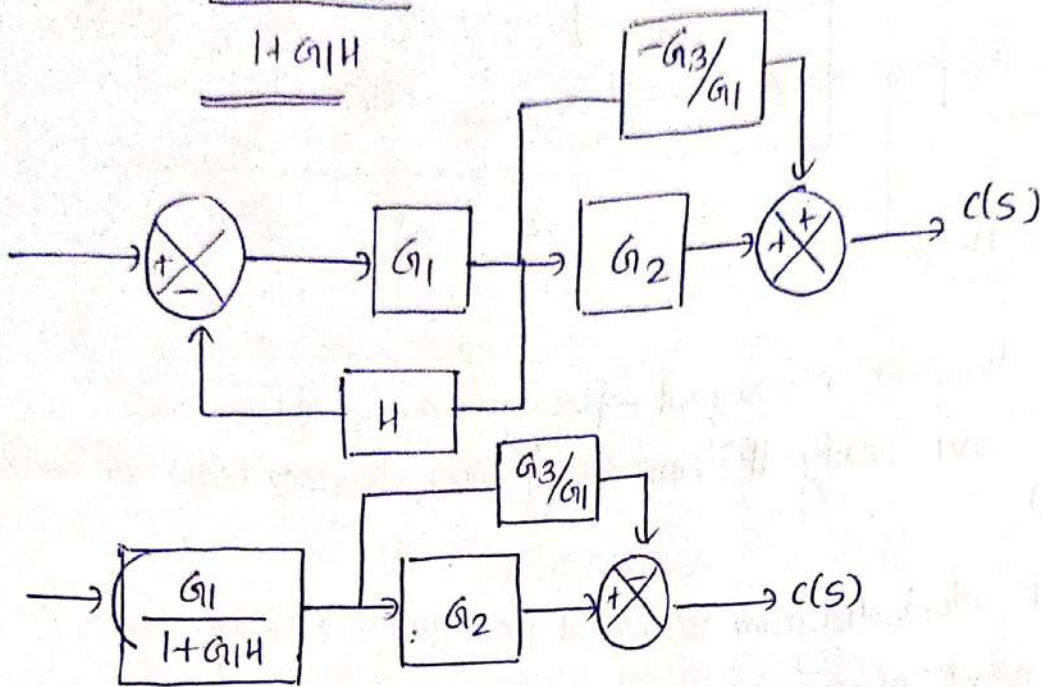
$$\Delta = 1 + G_1 H$$

$$\Delta_1 = 1$$

$$\Delta_2 = 1 + G_1 H$$

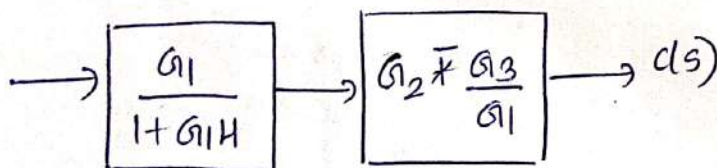
$$TF = \frac{G_1 G_2 - G_1 H (1 + G_1 H)}{1 + G_1 H}$$

$$TF = \frac{G_1 G_2 - G_3}{1 + G_1 H}$$



Wrong

$$\frac{C(s)}{R(s)} = \frac{G_1}{1+G_1H} \times \frac{G_1 G_2}{1+G_2 G_3}$$



$$\frac{G_1}{1+G_1H} \times \frac{G_2 G_1 - G_3}{G_1}$$

$$= \frac{G_1 G_2 - G_3}{1+G_1H}$$

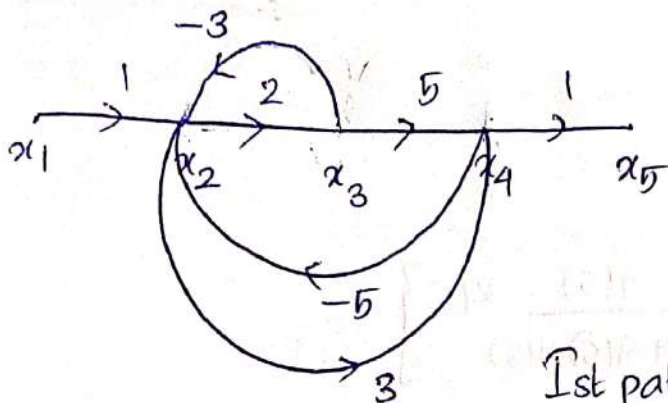
Qn: Find the gain of the system.

$$x_2 = x_1 - 3x_3 - 5x_4$$

$$x_3 = 2x_2$$

$$x_4 = 5x_3 + 3x_2$$

$$x_5 = x_4$$



1st path : $x_1 - x_2 - x_3 - x_4 - x_5$

2nd path : $x_1 - x_2 - x_4 - x_5$

$$P_1 = 10$$

$$P_2 = 3$$

$$L_1 : x_2 - x_3 - x_2 \quad P_{11} = -6$$

$$L_2 : x_2 - x_3 - x_4 - x_2 \quad P_{21} = -50$$

$$L_3 : x_2 - x_4 - x_2 \quad P_{31} = -15$$

$$\Delta = 1 - (-6 - 50 - 15)$$

$$= 1 + 71$$

$$= \underline{\underline{72}}$$

$$\Delta_1 = 1$$

$$\Delta_2 = 1$$

$$TF = \frac{10 + 3}{72} = \underline{\underline{\frac{13}{72}}}$$